Draft version July 29, 2026
Typeset using LaTeX twocolumn style in AASTeX7.0.2

MoCHII: A Three-Dimensional Monte Carlo Photoionization Code for Dusty Nebulae on Adaptive Octree Grids

Kwang-Il Seon [1,2]

[1] Korea Astronomy and Space Science Institute, 776 Daedeokdae-ro, Yuseong-gu, Daejeon 34055, Republic of Korea
[2] Department of Astronomy and Space Science, University of Science and Technology, Korea, 217 Gajeong-ro, Yuseong-gu, Daejeon 34113, Republic of Korea

ABSTRACT

We present MoCHII (MOnte Carlo for H II regions), a three-dimensional Monte Carlo radiative-transfer code for dusty photoionized nebulae on adaptive octree (AMR) grids. MoCHII is built by adding a self-consistent gas-physics layer to the validated octree transport, peel-off, and stochastic-dust modules of the dust code MoCafe, and it computes hydrogen and helium photoionization and thermal balance, a trace-metal ionization cascade with modern analytic atomic data, the dust and PAH continuum, and the recombination-line and nebular-continuum diagnostics all from one radiation field. The band transport uses an analytic (zero-variance) direct-field estimator on the octree together with a dedicated Cartesian digital-differential-analyzer (DDA) backend, an explicit Monte Carlo diffuse field, dust absorption and forward scattering, and observer peel-off imaging. The atomic data are fitted offline and evaluated online: exact hydrogenic H I/He II cross sections, Verner et al. (1996) fits for the remaining ions, Badnell (2023) radiative and dielectronic recombination with a Mao & Kaastra (2016) ground-level split for case A/B, Voronov collisional ionization, and charge exchange, so that adding a metal ion is a data operation rather than a code operation. We validate the code against analytic attenuation (rate-integral bias +0.004%), the Strömgren sphere (ionized volume +0.56% versus an exact 1D reference; AMR identical to a uniform grid), a thermal H/He nebula (median 0.14% in $T_{\rm e}$), and the Lexington 2000 H II region benchmarks. Its line cooling is taken from the level populations at the local electron density, the same n-level solve that produces the emergent lines, so the thermal balance and the line list are one physics. On the two benchmarks the $n_{\rm p}n_{\rm e}$-weighted temperature then agrees with a Cloudy c25.00 calculation of the same models to 0.2% (HII40) and 0.1% (HII20), with the $\rm He^+/H^+$ ratio and the outer radius inside the published multi-code ranges; the cooling itself is cross-validated channel by channel against a Cloudy c25.00 decomposition on a fixed $(T_{\rm e}, n_{\rm e}, \text{ionization})$ state, where it agrees to 1%. We further demonstrate solution-driven ionization-front re-refinement (a re-refined octree reproduces a native high-resolution grid to +0.008% in ionized volume with 4.5× fewer cells), the complete dust and PAH coupling (grain heating energy closure $L_{\rm IR}/L_{\rm abs} = 1.000$; the projected $8\,\mu$m image turning from a central peak into a limb-brightened ionization-front annulus as PAHs are destroyed in ionized gas), and an atomic photodissociation-region skin with photoelectric heating. MoCHII is written for modern three-dimensional synthetic observations of H II regions and their surroundings.

Keywords: H II regions (694) — Interstellar medium (847) — Radiative transfer (1335) — Photoionization (2060) — Interstellar dust (836) — Astronomy software (1855)

1. INTRODUCTION

Photoionized regions — H II regions, planetary nebulae, the diffuse ionized medium, and the circumgalactic gas illuminated by stars and active nuclei — are among the most information-rich objects in astrophysics. Their emission-line and continuum spectra encode the ionizing radiation field, the elemental abundances, the electron temperature and density, and the dust content of the gas. Extracting that information requires a radiative-transfer model that couples the ionizing radiation field

Email: kiseon@kasi.re.kr

to the ionization, thermal, and dust state of the gas self-consistently.

The classic tools for this problem are one-dimensional: Cloudy (G. J. Ferland et al. 2017; M. Chatzikos et al. 2023) and Mappings (R. S. Sutherland & M. A. Dopita 1993, 2017) solve the coupled ionization, thermal, and (in the case of Mappings) shock structure of a spherical or plane-parallel slab with very detailed microphysics: large ion inventories, density-dependent multilevel cooling, non-equilibrium and time-dependent options, and grain and PAH physics. Their strength is completeness of the local physics; their limitation is geometry. Real H ii regions are clumpy, aspherical, and shadowed, and the diffuse recombination field they produce is intrinsically three-dimensional.

Three-dimensional Monte Carlo photoionization codes were developed to address exactly this geometry. Mocassin (B. Ercolano et al. 2003, 2005, 2008) pioneered the self-consistent Monte Carlo treatment of the direct and diffuse fields on a Cartesian grid, with a large ion and dust inventory, and it defines the *de facto* three-dimensional benchmark suite (the Lexington/Meudon comparison cases; G. Ferland 1995; D. Péquignot et al. 2001). The code of K. Wood et al. (2004) demonstrated Monte Carlo photoionization with an efficient path-length estimator for the mean intensity. M^3 (Messenger Monte Carlo Mappings V; Y. Jin et al. 2022a,b) grafts three-dimensional Monte Carlo transport onto the extensive Mappings V microphysics, combining broad atomic physics with arbitrary Cartesian geometry. CMacIonize (B. Vandenbroucke & K. Wood 2018) adds a task-based, unstructured (Voronoi or adaptive Cartesian) grid and a modern software design. These codes share a common architectural constraint: their spatial grids are either uniform Cartesian or, in the CMacIonize case, unstructured, and their diffuse-field and grid-traversal designs are tightly coupled.

MoCHII (MOnte Carlo for H ii regions)[3] is a new three-dimensional Monte Carlo photoionization code built around a different design choice: it reuses the mature octree adaptive-mesh-refinement (AMR) engine, event-driven traversal, peel-off imaging, and stochastic dust/PAH emission of the dust radiative-transfer code MoCafe (Seon, in preparation)[4], and adds a gas-physics layer on top. The motivation is practical. Retrofitting an octree traversal onto an existing uniform-grid photoionization code is difficult: the ray-segment bookkeeping is entangled with the physics loop, and structural rewrites are error-prone (the Mocassin `pathSegment` traversal, for instance, carries a dual position representation with many mutation sites). An event-driven AMR transport engine, by contrast, already *is* the design such a retrofit would have to reach. MoCafe provides that engine, validated for dust transport, so MoCHII inherits octree AMR, an analytic direct-field estimator, a dedicated Cartesian digital-differential-analyzer (DDA) backend, peel-off synthetic imaging, MPI-3 shared-memory parallelism, and HDF5/FITS I/O without modifying the dust tree.

On top of that engine MoCHII adds: (i) a continuous-energy ionizing photon sampler, with a grouped threshold-aligned compatibility mode, and a zero-variance direct-field estimator; (ii) hydrogen and helium photoionization and thermal balance with modern recombination data; (iii) a trace-metal ionization cascade organized as a data-driven registry, so that adding an ion (currently eleven elements, C through Ca) is an offline data operation; (iv) an explicit Monte Carlo diffuse field with a case-A/B option; (v) the complete dust and PAH coupling in the ionizing and far-ultraviolet (FUV) bands, including equilibrium and stochastic dust emission; (vi) solution-driven ionization-front re-refinement of the octree; and (vii) recombination-line, collisional-line, and nebular-continuum diagnostics with leaf-resolved emissivity output for synthetic imaging. The atomic data are fitted offline from CHIANTI and the Strathclyde/Badnell tables and evaluated by analytic formulae at run time, with a provenance header on every coefficient file.

The remainder of this paper is organized as follows. Section 2 compares MoCHII with the established 1D and 3D codes. Section 3 describes the Monte Carlo transport, the grids, the diffuse field, the iteration and convergence scheme, and the quasi-random launch option. Section 4 documents the atomic data and the offline fitting pipeline. Section 5 presents the validation against analytic solutions and the Lexington benchmarks. Section 6 demonstrates the solution-driven I-front re-refinement and a cosmological post-processing application. Section 7 presents the dust and PAH coupling, and Section 8 the photodissociation-region skin. Section 9 summarizes and outlines planned extensions.

2. COMPARISON WITH EXISTING CODES

Table 1 places MoCHII next to the reference codes along the axes most relevant to three-dimensional nebular modeling. The comparison is deliberately architectural rather than a claim of superior physics: Cloudy and Mappings carry more complete local microphysics

[3] https://github.com/seoncafe/MoCHII

[4] https://github.com/seoncafe/MoCafe

(larger ion inventories, full multilevel cooling of every ion, non-equilibrium and shock options), and Mocassin carries a comparably broad ion and dust inventory in three dimensions. MoCHII's distinguishing features are the octree AMR grid with solution-driven re-refinement, the reuse of a validated dust-transport engine (so that dust absorption, forward scattering, equilibrium and stochastic emission, and PAH physics are first-class rather than bolted on), and the peel-off synthetic-imaging output.

The grid axis is the sharpest distinction. Mocassin and M^3 use uniform Cartesian grids; CMacIonize uses an unstructured Voronoi or adaptive Cartesian grid; MoCHII uses an octree whose refinement can be driven by the ionization solution itself (Section 6). On the Strömgren problem this lets MoCHII concentrate resolution at the ionization front and reproduce a native high-resolution run with several times fewer cells.

The diffuse-field axis separates the on-the-spot (OTS) approximation from explicit transport. MoCHII defaults to case-B OTS (no diffuse packets, consistent with the analytic direct-field estimator) but offers an explicit Monte Carlo diffuse field with case-A rates and ground-recombination emission, matching the Mocassin and Cloudy treatment; the benchmark runs of Section 5.2 use it, so that the transfer treatment is the same on both sides of that comparison.

The atomic-data axis reflects vintage. MoCHII uses exact hydrogenic cross sections, Verner et al. (1996) fits, Badnell (2023) recombination, CHIANTI v11 collision strengths, and Mao & Kaastra (2016) level-resolved radiative recombination — the same modern data that Cloudy uses — rather than the older compilations still carried by Mocassin.

The metal-treatment axis is the clearest algorithmic difference. Mocassin, M^3, CMacIonize, and the 1D codes solve a *full network*: the ionization of every element is one coupled nonlinear system in which the metal opacity consumes photons from the radiation field, the metal electrons enter n_e, and the metal photoheating enters the thermal balance, all iterated to self-consistency (radiation field $\leftrightarrow$ all ions $\leftrightarrow n_e \leftrightarrow T$). MoCHII instead adopts a *trace-species* treatment: because the metals are present at abundances $\sim 10^{-4}$–10^{-3}, they are separated as a perturbation on the H/He (and dust) skeleton — the radiation field and n_e are set by H, He, and dust, and each element's ionization cascade is an independent, effectively linear chain evaluated on the converged (J_ν, n_e, T). The one feedback that is *not* dropped is metal line cooling, which enters the thermal balance because the H II-region temperature is itself largely the result of metal cooling (Section 4.4). Where the trace separation genuinely breaks down, it is restored selectively by switches (the default is trace, with recoupling only where the physics demands it): `ion_metal_abs` adds the metal photoionization opacity to the band (on by default), `metal_ne` feeds the metal electrons into the n_e closure — essential in neutral photodissociation-region gas, where carbon dominates the electron budget, and which the calculation of Section 8 uses — and `metal_heat` adds metal photoheating to the thermal balance. The design payoff is that, with each element an independent linear chain, adding an ion is a data-file addition rather than a change to a coupled solver (the registry of Section 4), and the cell solve stays at the level of closed-form evaluations rather than a coupled-network iteration.

A related but separate axis is the cooling inventory — which channels a code carries at all, since that is what sets the equilibrium temperature. As implemented in the distributed source, the Monte Carlo code of K. Wood et al. (2004) cools by ten five-level metal ions (N I, N II, O I, O II, O III, Ne III, S II, S III, C II, C III) plus two two-level ions ([N III] 57 μm and [Ne II] 12.8 μm), and CMacIonize by the same ten five-level ions plus three two-level ions, adding [S IV] 10.5 μm; both then add free-free emission and hydrogen and helium recombination cooling, in each case with the H^+ and He^+ terms only. MoCHII carries line cooling for 28 ions — H I and the 27 metal ions of the eleven-element registry (Appendix A) — and adds three channels that neither of those two codes carries: H I collisional excitation, collisional-ionization cooling of H and He, and the He^{++} terms of the free-free and recombination sums. In a 10^4 K H II region these are small (H I collisional lines are 0.7% of the HII40 cooling budget), but they are the terms that take over in the hot, partly neutral gas of an ionization front and in the shielded gas beyond it.

3. METHODS

MoCHII follows the conventions of its parent dust code: all input is a single Fortran namelist (`par`); computation is in double precision; parallelism is MPI-only with MPI-3 shared-memory windows for the grid; output is HDF5 with a FITS fallback; and algorithm variants are selected by procedure pointers at startup. A run reads (or builds) a grid, persists a gas leaf state, and iterates transport and equilibrium to convergence.

3.1. *The ionizing band and the source*

The ionizing band spans photon energies $[E_{\min}, E_{\max}] = [13.598, 100]$ eV. In continuous mode, the source spectrum — a Planck function $B_\nu(T_\star)$ or a tabulated spectrum in shape or physical (per-eV, per-Hz,

Table 1. Architectural comparison of MoCHII with reference photoionization codes. "OTS" = on-the-spot (case B); "explicit" = transported diffuse field. This table compares design, not the breadth of local microphysics, on which the 1D codes and MOCASSIN are more complete.

	MoCHII	MOCASSIN	M^3	CMacIonize	Cloudy / MAPPINGS
Dimensionality	3D	3D	3D	3D	1D
Grid	octree AMR + Cartesian DDA	uniform Cartesian	uniform Cartesian	Voronoi / adaptive Cartesian	spherical / plane-parallel
Solution-driven refinement	yes (I-front)	no	no	static AMR	1D shells
Diffuse field	OTS default; explicit MC option	explicit MC	explicit MC	explicit MC	outward-only / OTS
Recombination data	Badnell 2023 + Mao & Kaastra 2016	Verner & Ferland 1996	MAPPINGS V	Verner / Badnell	Badnell / internal
Metal treatment	trace-species registry (data-driven)	full network	full MAPPINGS network	full network	full network
Dust / PAH	absorption + forward scattering + equilibrium + stochastic + PAH survival	absorption + emission	MAPPINGS grains	absorption + emission	detailed grains/PAH
Parallelism	MPI-3 shared memory	MPI	MPI	task-based / MPI	OpenMP / serial
Synthetic output	peel-off images, line maps, IR SED	line maps, images	line maps	line maps, images	1D spectra

per-Å, per-μm) units — defines an energy-luminosity probability density. A packet energy is drawn by inversion of its CDF and the photoionization cross sections and excess energy are evaluated at that drawn energy. Thus the n_ν bins (typically 32) are a diagnostic tally of J_E, not a transport discretization; changing their count cannot change the physical solution. Packets carry $L_{\rm pkt} = L_{\rm band}/N$.

Grouped mode.—The grouped mode (`ion_energy_mode = 'grouped'`) samples a bin from the luminosity CDF and uses its representative energy. In that mode, by default (`ion_align_edges`) the bin *edges* are pinned onto the ionization thresholds — H I (13.6 eV), He I (24.6 eV), He II (54.4 eV), and the active metal edges gathered from the registry — so that no bin straddles a threshold; the sub-segments between thresholds are log-uniform. This removes a real discretization error: a coarse 32-bin log grid straddles the 24.6 eV He I edge and delivers ∼18% too many He-ionizing photons against a true 40 kK blackbody ($Q_{\rm HeI}/Q_{\rm H} = 0.127$ versus the exact 0.108), over-ionizing helium and, downstream, the metals. Aligning the edges restores the exact ratio at fixed bin count; this is the single largest lever on the He and metal ionization structure in a grouped HII40 calculation. Continuous mode has no such bin-edge dependence. When thresholds outnumber bins the lowest-priority metal edges are dropped gracefully, so the option is safe at any bin count, and disabling it recovers the plain logarithmic grid bit-for-bit.

FUV extension and composite sources.—With `par%add_fuv` the band grows a second, lower segment of FUV bins on $[E_{\rm FUV}, E_{\rm min}]$ (default $E_{\rm FUV} = 6$ eV) below the unchanged ionizing bins. H and He cross sections vanish below their thresholds, so the FUV bins interact only with dust and with the low-threshold metal stages, producing the neutral-zone He I-like species (e.g. Mg II, C II) that the ionizing-only band cannot. The band is fed by any number of internal point sources (0–16) plus, independently, one isotropic external radiation field. The external field is a band-integrated mean intensity J [$\rm erg\,s^{-1}\,cm^{-2}\,sr^{-1}$] entering through the box faces or a bounding sphere; the inward flux is the hemisphere integral of $J\cos\theta$, so the entering band power is $L = \pi J A_{\rm surf}$, and entry directions are cosine-weighted (Lambert), reproducing an isotropic interior field of energy density $u = 4\pi J/c$. Three analytic interstellar-radiation-field presets (Draine, Habing, Mathis) are available for the FUV segment.

3.2. *Composable sources: a cloud in an external field*

The source composition described above combines any number of internal point sources with one external field, so a cloud can be modeled as it is actually illuminated. Figure 1 is the limiting case of no internal star: a uniform sphere ($n_{\rm H} = 1.7\,{\rm cm}^{-3}$, $R = 1.8$ pc) bathed in an isotropic external field with a 4×10^4 K blackbody shape, ionized entirely from outside. The resulting structure inverts that of a central-source Strömgren sphere. An ionized skin ($x_{\rm HII} = 0.93$ in the outer shell) wraps a more neutral core (median $x_{\rm HI} = 0.94$), and the ionized-to-neutral crossover falls near $r = 1.05$ pc. The planar slice through the center and the shell-averaged radial profiles both show the field working inward from the illuminated surface. A cloud embedded in an interstellar or circumgalactic radiation field is therefore computed directly, without imposing the incident field as a boundary condition on a separate one-dimensional model.

3.3. *Opacity and the zero-variance edge walk*

Each leaf carries the band absorption coefficient

$$\kappa_b = n_{\rm H}\big[x_{\rm HI}\,\sigma_{\rm HI}(E_b) + y_{\rm He}\big(x_{\rm HeI}\sigma_{\rm HeI} + x_{\rm HeII}\sigma_{\rm HeII}\big)(E_b)\big]\,d_{\rm cm}\;(+\,\kappa_{\rm dust}), \quad (1)$$

stored as the full $(n_\nu, n_{\rm leaf})$ array, plus optional metal photoionization opacity. Because the ionizing band has no scattering (without dust) and the diffuse field, when used, is emitted explicitly, each packet's contribution to the mean-intensity tally is computed *analytically* along its ray — the forced-first-flight estimator of L. B. Lucy (1999), made exact:

$$\Delta J_b({\rm leaf}) = w\,L_{\rm pkt}\,\frac{e^{-\tau_{\rm in}} - e^{-\tau_{\rm out}}}{\kappa_b}, \quad (2)$$

accumulated leaf by leaf until the domain edge. No interaction sampling is needed; the only Monte Carlo variance is in the direction and bin sampling. From the reduced tally J_b the code forms, for each absorber i, the photoionization rate and photoheating rate

$$\Gamma_i = \sum_b \frac{[4\pi J\,\Delta\nu]_b\,\sigma_i(E_b)}{h\nu_b},$$

$$\mathcal{H}_i = \sum_b [4\pi J\,\Delta\nu]_b\,\sigma_i(E_b)\left(1 - \frac{E_{{\rm th},i}}{E_b}\right). \quad (3)$$

3.4. *Grids: octree AMR and the Cartesian DDA backend*

MoCHII inherits the MoCafe octree. The grid may be read as an AMR leaf list, built from a namelist size $(n_x\times n_y\times n_z)$ with an optional uniform-density sphere or shell, or read from a 3D density cube. Geometry is fully analytic behind accessor functions, so a leaf's center and half-size are array reads for a general octree and raster-index decodes for a uniform Cartesian ("car") grid — the same floating-point expressions in both cases. The car backend allocates no tree or neighbor arrays at all (a saving that scales to $\sim$10 GB at 512^3) and traverses rays with a dedicated incremental Amanatides–Woo DDA walk, which is the default for uniform grids. A verification mode reproduces the single-level octree traversal bit-for-bit, and the DDA agrees with it to rounding, at up to twice the transport throughput. The two backends share all of the physics, transport, and peel-off code.

3.5. *Ionization and thermal equilibrium*

In each leaf, the ionization state is solved (`solve_ion_cell`) from the ionization rates $R_i = \Gamma_i + n_{\rm e}C_i(T)$ and the case-A/B recombination rates α_i: $x_{\rm HI} = \alpha n_{\rm e}/(R_{\rm HI} + \alpha n_{\rm e})$, the He chain $n_{i+1}/n_i = R_i/(\alpha_{i+1}n_{\rm e})$, closed by $n_{\rm e} = n_{\rm H}[x_{\rm HII} + y_{\rm He}(x_{\rm HeII} + 2x_{\rm HeIII})]$ and solved by a damped fixed point on $n_{\rm e}$. When thermal balance is enabled, the temperature is updated once each iteration by finding the root of the net rate $\mathcal{N}(T) = \mathcal{H}(T) - \mathcal{C}(T)$ by bisection on $\log T_{\rm e}$; the ionization state is re-solved at every trial T. The cooling $\mathcal{C}(T)$ assembles recombination cooling, free-free (with the net ion charge Z, including the trace metals at $Z_{\rm ion} = {\rm stage} - 1$), collisional ionization, H I line cooling, and the trace-metal cooling sum $\sum n_{\rm e}n_{\rm ion}\Lambda_{\rm ion}(T)$. Optional secondary ionization by fast photoelectrons (J. M. Shull & M. E. van Steenberg 1985) redistributes the photoelectron energy between heating and secondary H I/He I ionization in partially neutral gas.

3.6. *The iteration loop and convergence*

Each iteration zeroes the band tally, transports the stellar packets (and, if enabled, rebuilds and transports the diffuse packets), reduces the tally over MPI, forms the rate integrals, solves each leaf, and refills the opacity from the new state; the stellar tally is recomputed from zero each iteration. Every rank runs the identical cell solves from the reduced tallies, so all ranks agree on convergence, and only the node master writes the shared state.

Convergence is measured two ways every iteration, and `par%conv_crit` selects which pair controls the loop: the largest change found in any single cell ($\max|\Delta x_{\rm HII}|$, $\max|\Delta T_{\rm e}|/T_{\rm e}$) and the volume-integrated measures

$$\frac{\sum_{\rm leaf} V\,|\Delta x_{\rm HII}|}{\sum_{\rm leaf} V\,x_{\rm HII}}, \qquad \frac{\sum_{\rm leaf} V\,n_{\rm e}\,|\Delta T_{\rm e}|}{\sum_{\rm leaf} V\,n_{\rm e}\,T_{\rm e}}. \quad (4)$$

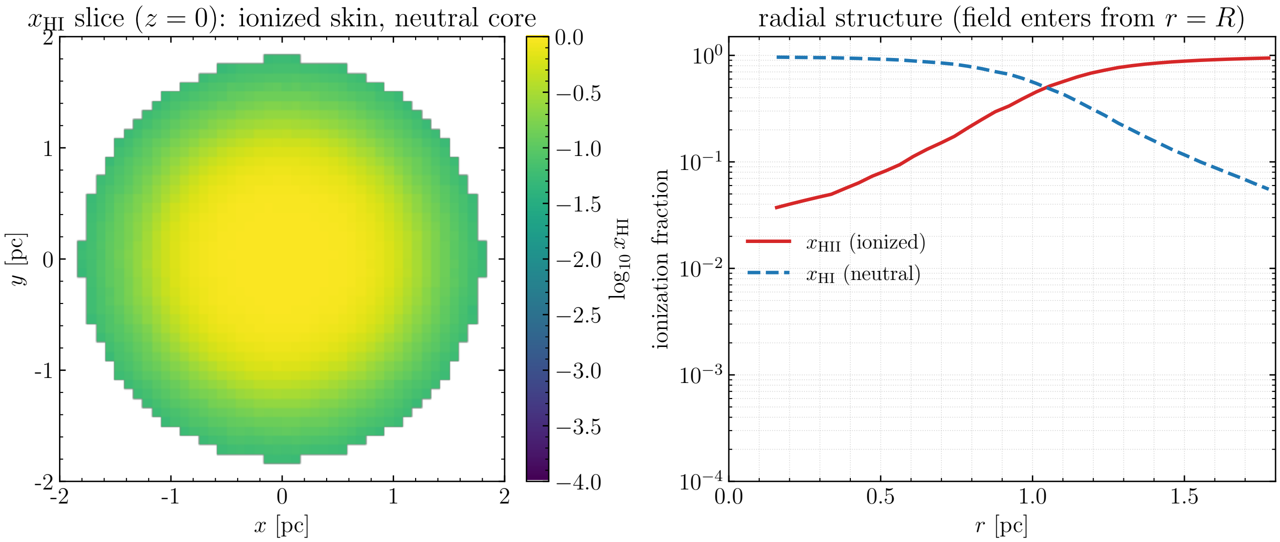

Figure 1. A uniform cloud ($n_{\rm H} = 1.7\,{\rm cm}^{-3}$, $R = 1.8$ pc) photoionized by an isotropic external radiation field entering the box faces (a 4×10^4 K blackbody shape), with no internal source. Left: a planar slice of the neutral fraction $x_{\rm HI}$ through the cloud center, showing an ionized skin over a neutral core. Right: the shell-averaged radial profiles of $x_{\rm HI}$ and $x_{\rm HII}$, with the field entering at the surface $r = R$. The ionization structure is the inverse of that produced by a central source.

The volume measures fix two stall modes of these single-cell maxima: ionization-front jitter occupies little volume (the front shell is a few thousand cells against $\sim 10^5$ ionized ones, so its Monte Carlo noise enters the L_1 norm suppressed by that ratio), and the $n_{\rm e}$ weight removes the vacuum and no-heating cells whose floor-pinned oscillation dominates $\max |\Delta T_{\rm e}|$ in FUV/PDR configurations. On a FUV dusty Strömgren case at 8×10^6 packets, where these single-cell maxima stall at 0.1 and 6.4, the volume measures decay smoothly to Monte Carlo floors of 2.7×10^{-3} and 6.6×10^{-3} by iteration $\sim$27, and 30 further iterations leave the ionized volume unchanged to 0.014%. The floors scale as $1/\sqrt{N_{\rm packets}}$, and the volume criterion is the recommended production setting.

3.7. *Explicit diffuse field*

When the explicit diffuse field is enabled, case-B OTS is replaced by case-A rates plus explicit emission of ground-recombination packets in three channels (H II, He II, He III) with rate $\alpha_1 = \alpha_A - \alpha_B$ and photon energies $E = E_{\rm th} + kT_e\, x$, $x \sim {\rm Exp}(1)$. The emission table is rebuilt from the current state each iteration; packets are transported by the same edge walk. The sampled photon energy is mapped to its bin by a binary search of the stored bin-edge array, correct on both the aligned and the plain logarithmic grids: an energy exactly on a threshold edge falls in the bin *above* it, the physically correct side for a recombination photon emitted just above threshold. An optional fourth channel adds the He I excited-level recombination radiation with the low-density branching of D. E. Osterbrock & G. J. Ferland (2006). At HII40 the diffuse band luminosity converges to $\sim 0.5\, L_\star$.

3.8. *Quasi-random photon launching*

Optionally the packet launch draws its variables (source component, photon energy through the source CDF in continuous mode or frequency bin in grouped mode, direction, and any entry-surface coordinates) from an Owen-scrambled Sobol sequence (S. Joe & F. Y. Kuo 2008; B. Burley 2020) instead of the Mersenne Twister, while every post-launch transport event stays pseudo-random. The Sobol point is addressed by the *global* photon number, so the launch set is independent of the MPI task count. Because the ionizing band has no scattering, the direct field is an analytic function of the launch variables (Eq. 2), and a low-discrepancy launch set reduces its noise directly.

Figure 2 shows the resulting scaling on the fixed-state analytic-attenuation test. The Monte Carlo cell-wise root-mean-square error in $\Gamma_{\rm HI}$ follows the $N^{-1/2}$ reference slope while the Owen-scrambled Sobol launch follows N^{-1}, so the effective photon gain, the ratio of the two variances, grows with the packet count: 28.8× at $N = 2^{17}$, 66.5× at 2^{18}, 141× at 2^{19}, and 222.5× at 2^{20}. The two launch sets cost the same wall time, and the quasi-random result carries no significant bias.

3.9. *Peel-off imaging*

After convergence, one extra transport pass runs with observer peeling (the MoCafe convention adapted to the band). Every emitted packet peels a direct contribu-

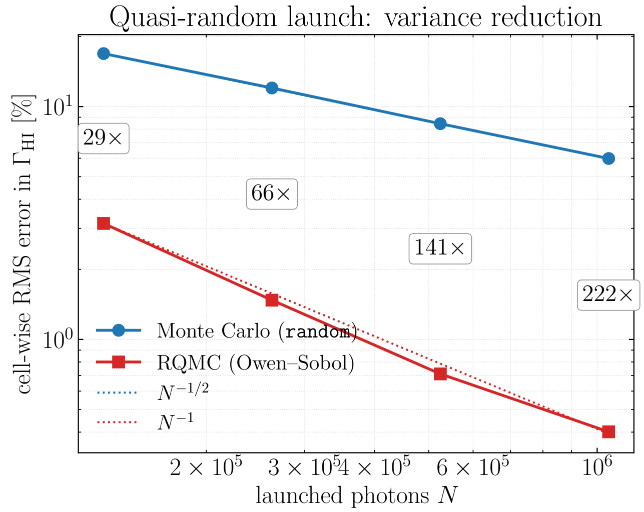

Figure 2. Quasi-random photon launching. Cell-wise root-mean-square error in $\Gamma_{\rm HI}$ on the fixed-state analytic-attenuation test, as a function of the launched photon count N, for the ordinary Monte Carlo (`random`) and Owen-scrambled Sobol (RQMC) launches, with the $N^{-1/2}$ and N^{-1} reference slopes. The boxed labels give the effective-photon gain (the variance ratio) at each N, which grows with N as the two slopes diverge.

tion $L_{\rm pkt}\, w\, e^{-\tau}/(4\pi r^2)$ toward each observer, and, when dust scattering is sampled, every interaction peels a scattered contribution with the Henyey–Greenstein phase function. The optical depth to the box edge is integrated by a tally-free walk, so the images carry exactly the transport's extinction. Images are TAN projections written as surface brightness, split into direct/scattered and ionizing/FUV channels; with a spectral cube option the channel axis becomes the band bin for hardness maps. Leaf-resolved line and continuum emissivities are also written, from which flux-conserving line and field maps are produced.

Figure 3 is an example: the FUV channel of a dusty, radiation-bounded H II region with a central star, seen by the observer. The direct image is the attenuated stellar point source (peak $\log_{10} I = 1.95$ in $\mathrm{erg\,s^{-1}\,cm^{-2}\,sr^{-1}}$), and the scattered image is the dust halo it illuminates (peak 0.24, covering some 27% of the pixels). The halo is the observer-dependent component that a one-dimensional or angle-averaged calculation does not provide, and the sum of the two is the surface brightness an observer measures. The ionizing band contributes almost nothing to the scattered image here, because in a bounded nebula it is absorbed before it can scatter out. Alongside the extincted image the code also writes the bin-resolved image cube and an unattenuated (`direc0`) image, the latter giving a line-of-sight optical-depth map as its ratio to the extincted one.

4. ATOMIC DATA

The design principle for the atomic data is that adding an ion is a *data operation, not a code operation.* All rate coefficients are fitted offline by a Python pipeline from CHIANTI v11.0.2 (K. P. Dere et al. 1997; R. P. Dufresne et al. 2024) and the Strathclyde/Badnell tables, and are evaluated online by fixed analytic formulae. Every coefficient file carries a provenance header (source, version, fit range, maximum fit error), so the coefficients are auditable and the Fortran only evaluates formulae.

4.1. *Photoionization cross sections*

The one-electron ions H I ($Z=1$) and He II ($Z=2$) use the exact ground-state hydrogenic cross section (D. E. Osterbrock & G. J. Ferland 2006)

$$\varepsilon = \sqrt{E/E_{\rm th}-1},$$
$$\sigma(E) = \frac{\sigma_0}{Z^2}\left(\frac{E_{\rm th}}{E}\right)^4 \frac{\exp\!\big(4-4\arctan\varepsilon/\varepsilon\big)}{1-e^{-2\pi/\varepsilon}}, \qquad (5)$$

with $\sigma_0 = 6.30\times10^{-18}\,\mathrm{cm^2}$ and $E_{\rm th}$ the true ionization potential (13.598 and 54.416 eV), tied to the physical edge so it stays aligned with the band grid. He I and every metal ion use the single-shell analytic fit of D. A. Verner et al. (1996):

$$x = \frac{E}{E_0} - y_0, \qquad z = \sqrt{x^2+y_1^2},$$
$$\sigma(E) = \sigma_0\left[(x-1)^2+y_w^2\right] z^{-Q}\left(1+\sqrt{z/y_a}\right)^{-P},$$
$$Q = 5.5 - \tfrac{1}{2}P, \qquad (6)$$

with parameters parsed directly from the authors' `phfit2.f` table. For the elements absent from that table (P, Cl, K — Opacity-Project gaps), the D. A. Verner & D. G. Yakovlev (1995) outer-shell subshell fit is used instead, the same hybrid as Mocassin. The cross sections are cross-checked against the independent implementation in the exoplanet-atmosphere code EXHALE.

4.2. *Recombination*

The default recombination model takes the total (case-A) radiative recombination α_A from the N. R. Badnell (2006) radiative recombination fit,

$$\alpha_{\rm RR}(T) = \frac{A}{\sqrt{T/T_0}\big(1+\sqrt{T/T_0}\big)^{1-b}\big(1+\sqrt{T/T_1}\big)^{1+b}},$$
$$b = B + C\,e^{-T_2/T}, \qquad (7)$$

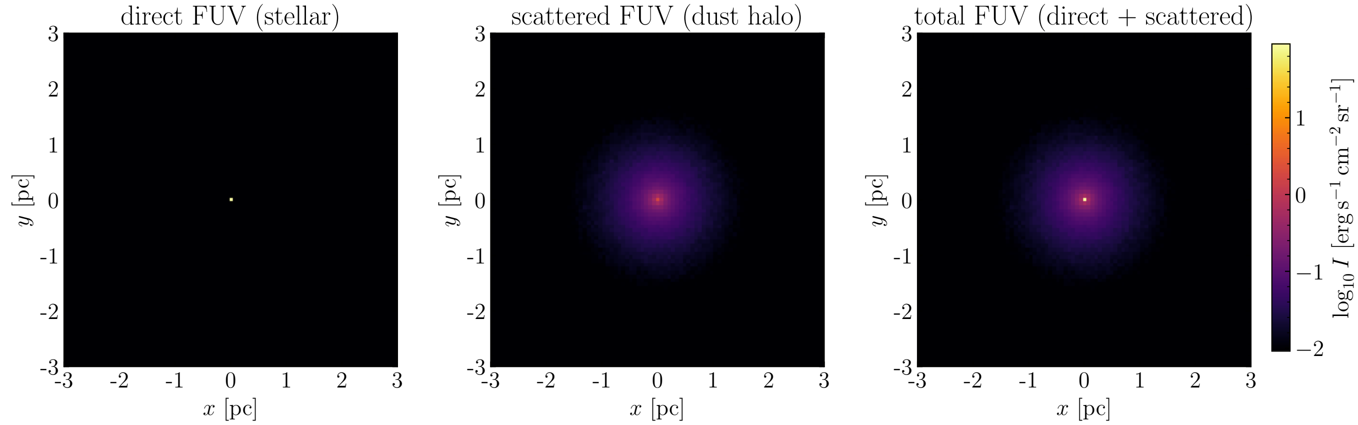

Figure 3. Peel-off synthetic imaging of a dusty, FUV-illuminated H II region with a central star, in the FUV channel. Left: the direct stellar image, an attenuated point source. Center: the dust-scattered halo, the observer-oriented contribution. Right: their sum, the synthetic observation. All three panels share one logarithmic scale of the specific intensity I integrated over the channel's band, so the extended scattered halo is directly comparable to the compact direct source.

the ground-level ($n = 1$) direct-capture rate α_1 from the level-resolved parameterization of J. Mao & J. Kaastra (2016), and forms case B by removing the ground term, $\alpha_B = \alpha_A - \alpha_1$. For the H-like ions this α_1 is derived from the exact hydrogenic cross sections (Eq. 5) under detailed balance, so it is exact; the $He\,\textsc{ii} \to He\,\textsc{i}$ channel adds the Badnell three-term dielectronic sum. This construction mirrors how CLOUDY splits case A and case B (total minus resolved ground) and reuses the same modern data. An alternative setting selects the L. Hui & N. Y. Gnedin (1997) fits; the two agree to <0.6% for H II and He III, the one substantive difference being $He\,\textsc{ii} \to He\,\textsc{i}$, where the Badnell value ($4.37 \times 10^{-13}\,\mathrm{cm^3\,s^{-1}}$ at 10^4 K) is ∼7% higher and coincides with CLOUDY, while the older L. Hui & N. Y. Gnedin (1997) value is low and the MOCASSIN Verner & Ferland value (4.84) is the high outlier. For the metal cascade, total recombination is the N. R. Badnell (2006) radiative recombination plus the Badnell-group dielectronic fits $\alpha_{\rm DR}(T) = T^{-3/2} \sum_i c_i e^{-E_i/T}$ (N. R. Badnell et al. 2003), read from the Strathclyde tables (verified md5-identical to the copies CLOUDY redistributes), with a CHIANTI fallback for the few ions Badnell does not tabulate.

4.3. *Collisional ionization and charge exchange*

Collisional ionization uses the G. S. Voronov (1997) fit

$$k(T) = A\,\frac{1 + P\sqrt{U}}{X + U}\,U^K e^{-U}, \qquad U = \frac{\Delta E}{kT}, \tag{8}$$

with an optional hybrid that multiplies each rate by the constant K. P. Dere (2007)/Voronov ratio of that element and stage; the difference matters only in shielded, low-ionization zones where collisions are the sole metal-ionization channel. Charge exchange with hydrogen uses C. Huang et al. (2023) Table 4 for the first stages and the J. B. Kingdon & G. J. Ferland (1996) fits $k = a\,T_4^b(1 + c\,e^{d\,T_4})$ for the higher stages (recombination direction). The tabulated higher-stage rates are indexed by the *product* ion of the reaction $\mathrm{X}^{i+} + \mathrm{H}^0 \to \mathrm{X}^{(i-1)+}$, a convention worth stating explicitly because it is easy to misread by one ionization stage and because the doubly and triply ionized metal fractions are sensitive to it; the indexing used here is checked against MOCASSIN's ionization loop and CMACIONIZE.

4.4. *Cooling and n-level diagnostic data*

Collisional line cooling enters the code at two levels, both built from the same CHIANTI v11.0.2 model of each ion (level energies from the observed term structure, radiative A-values, and effective collision strengths descaled after A. Burgess & J. A. Tully 1992). The thermal loop needs a cooling rate at every trial temperature of the bisection, and therefore uses a closed-form fit suppressed to the local density (below); the diagnostic output needs density-dependent emissivities, and therefore solves the level populations. Both are the same statistical-equilibrium solution of each ion, so the thermal balance and the emergent line list are one physics.

The thermal-loop coefficient is a multi-exponential fit

$$\Lambda_{\rm ion}(T) = T^{-1/2} \sum_{i=1}^{N} A_i\,e^{-T_i/T} \quad [\mathrm{erg\,cm^3\,s^{-1}}], \tag{9}$$

a rate coefficient multiplying $n_{\rm e} n_{\rm ion}$, fitted over 10^3–10^5 K to the $n_{\rm e} \to 0$ limit of that ion's own statistical-equilibrium solution — all population in the lowest level, excitation at the observed level energies, and the full cascade energy radiated. The maximum fit error over the temperatures that carry the cooling is 2.2% across the default C/N/O/Ne/S set (the coefficients of every ion are distributed with the code, Appendix A; the heavily depleted Si/Cl/Ca ions have larger full-range fit errors, confined to the temperature tails where the cooling contribution is negligible).

By itself Eq. (9) carries no collisional de-excitation, so it is exact for lines whose critical density is far above the local n_e and increasingly too large for those below it — the far-infrared fine-structure transitions [O III] 52 and 88 μm and [C II] 158 μm, whose electron critical densities are 10^2–$10^{3.5}$ cm^{-3}, so at the $n_e \sim 10^2$ cm^{-3} of an H II region it overestimates the line-cooling budget by about 12% (Section 5.3). The thermal loop therefore suppresses the coefficient to the local density. A (T_e, n_e) table, built once at setup by solving the same n-level atom over a grid and stored as the ratio $\Lambda_{SE}(T_e, n_e)/\Lambda_{SE}(T_e, n_e \to 0)$, multiplies Eq. (9) at each trial temperature; the fraction of the low-density cooling the (often truncated) n-level model does not resolve — its high-excitation resonance lines, of far higher critical density — is left unsuppressed, and as $n_e \to 0$ the suppression is unity and the coefficient returns to the fit exactly. Every ion carrying a cooling fit also carries an n-level model, so none is left in the zero-density limit; the two neutrals C I and N I are the most strongly suppressed of all, their far-infrared fine-structure lines having n_{crit} of only a few cm^{-3}. This density-suppressed cooling (`cooling_model = local_ne`, the default) is built from the very solve that produces the diagnostic lines, so the two are one physics; the pure zero-density fit is retained as the `low_density` option, exact in the diffuse gas the limit is written for, and the planetary-nebula densities of Section 9 require the table in any case.

For the output-time diagnostics, the lowest n levels, radiative A-values, and effective collision strengths $\Upsilon(T)$ (low-order Chebyshev fits in Burgess–Tully scaled space; worst transition 1.7%, Table 3) are exported for the line-emitting ions of the eleven-element registry. A statistical-equilibrium solve at output time (`nlevel_mod`) uses these to compute the collisional emission-line luminosities exactly — the density-dependent emissivities the low-density cooling fits cannot provide. The fitted files reproduce direct CHIANTI spline evaluations of the classic diagnostic ratios to 0.05–0.5%, and the Fortran solver matches an independent Python reference solver to four significant digits. PyNeb cross-checks give [O III] 5007/4363 to 1.2% and [N II] 6584/5755 to 0.3%; the density diagnostics and the iron lines differ by up to 6–16% from PyNeb's independent collision data — the well-known database spread, a data-vintage bracket rather than a code error.

4.5. *Recombination lines and the nebular continuum*

Hydrogen and He II recombination-line emissivities are interpolated from the P. J. Storey & D. G. Hummer (1995) case-A/B tables (following `par%case_ab`), and He I lines from the R. L. Porter et al. (2012, 2013) case-B collisional-radiative emissivities. The nebular continuum combines the free-bound plus free-free coefficients of the MOCASSIN (B. Ercolano & P. J. Storey 2006) tables with a Milne extension above the edges and the three two-photon continua; the free-free Gaunt factor is the D. G. Hummer (1988) fit, with the P. A. M. van Hoof et al. (2014) table as a drop-in option (the two agree to a median 5×10^{-5} in T_e).

4.6. *The He I 2^3S metastable diagnostic*

The metastable 2^3S level of neutral helium is the lower level of the 10833 Å triplet, whose absorption traces the extended atmospheres of close-in exoplanets and the neutral gas of nebulae. In H II regions the triplet is seen in emission, and near-infrared spectrographs such as NIRSPEC, CARMENES, and X-Shooter resolve its profile in Galactic regions like M17-B and NGC 3603 and in extragalactic star-forming regions (B. T. Draine 2026). MoCHII solves its population leaf by leaf on the converged state, following the balance of A. Oklopčić & C. M. Hirata (2018): recombination into the triplet ladder and collisional feeding from the ground state populate 2^3S, while radiative decay, collisional de-excitation, Penning ionization by neutral hydrogen, and photoionization by the transported radiation field depopulate it. From the resulting density the line-center optical depth of the 10833 Å doublet is formed. The diagnostic is pure post-processing, with no feedback on the ionization or thermal balance.

Figure 4 shows both sides of the transition for a dense compact example ($n_H = 10^4$ cm^{-3}, $R = 0.15$ pc, T_e fixed at 10^4 K). On the absorption side, the 2^3S column density reaches $N = 3.4 \times 10^{15}$ cm^{-2} and the line-center optical depth of the blended $J' = 1, 2$ doublet reaches 4.1×10^3 on the central sightline, falling through $\tau_0 = 1$ only at a projected radius of 0.139 pc, so all but the outermost rim of the projected disk is optically thick at 10833 Å. Such a column makes the line resonantly scatter, as B. T. Draine (2026) shows for H II regions: repeated scattering broadens the emergent profile into a multipeaked feature whose width can exceed 100 km s^{-1} and whose centroid is blueshifted by up to $\sim$14 km s^{-1} relative to other lines, while the extended path length lets dust absorb a larger share of the photons and so weakens the line. The He I 10833/Paγ ratio therefore stops being a clean He$^+$/H$^+$ measure once the metastable column is large. The optically-thin diagnostic likewise underestimates the true absorption over most of the region, which is what makes the 2^3S output of Figure 4 the input to a scattering calculation rather than an observable in itself. That calculation is within reach: the converged 2^3S density is handed to LaRT (K.-i. Seon & C.-G. Kim 2020; K.-i. Seon et al. 2022; K.-i. Seon 2024), a Monte Carlo resonant-line transfer code that already carries a 10833 Å module and has been applied to the line in an exoplanet atmosphere (D. Yan et al. 2022). The coupling is in place — a converter writes the MoCHII metastable grid in LaRT's input format, and the two codes agree on the line-center optical depth of the example above to 0.5% — and the resonant-scattering profiles will be presented in a forthcoming paper. On the emission side, the same transition radiates the 10833 Å line, and the code writes its emissivity from the R. L. Porter et al. (2012) recombination table — a collisional-radiative total, so it is used as tabulated and nothing is added to it. Integrated, the line carries $1.34\,L(\mathrm{H}\beta)$ here, and its emissivity falls at $r \simeq 0.13$ pc, inside the ionized sphere: the He^+ zone ends where the >24.6-eV photons run out, while hydrogen stays ionized to 0.15 pc.

5. VALIDATION

MoCHII is validated against analytic solutions, an exact 1D reference solver, and the Lexington benchmark suite, in staged tests. Each AMR result is required to reproduce the equivalent uniform-grid run.

5.1. *Rate integrals and the Strömgren sphere*

The zero-variance edge walk (Eq. 2) is verified on a uniform sphere ($n_\mathrm{H} = 1$, $R = 1$ pc, $x_\mathrm{HI} = 0.1$) with a central 10^5 K source. Against the cell-volume-averaged analytic attenuation, Γ_HI has a mean bias of +0.004% and a median deviation of 0.38% (pure Monte Carlo noise), and Γ_HeI a bias of +0.011% (Figure 5).

The Strömgren test uses a uniform $n_\mathrm{H} = 100$ sphere, $Q_\mathrm{H} = 10^{49}\,\mathrm{s}^{-1}$ ($T_\star = 4\times10^4$ K), case B, and $T_\mathrm{e} = 10^4$ K. For the reference we solve the spherically symmetric ionization-equilibrium equations directly, with a deterministic radial marching over 2×10^4 shells that carries the same band bins, photoionization cross sections, and recombination and collisional rates as the code: each bin's photon flux is attenuated with the running optical depth evaluated at mid-shell, and the H/He ionization equilibrium is solved shell by shell. The sweeps must update opacities Gauss–Seidel style (a Jacobi sweep, using the previous sweep's opacities, advances the front only one absorption length in each pass); a few sweeps then converge. "Exact" means free of Monte Carlo noise and grid-converged at this resolution — in the sharp-edge limit it reproduces the analytic Strömgren radius $R_S = [3Q_\mathrm{H}/(4\pi n_\mathrm{H}^2 \alpha_B)]^{1/3} = 3.153$ pc, from which the full solution differs physically (helium consumes photons; the front has a finite width). The 1D reference gives an effective ionized radius $R_\mathrm{eff} = (3V_\mathrm{ion}/4\pi)^{1/3} = 3.0365$ pc; a uniform level-6 octree gives 3.0534 pc (+0.56%), and a front-refined octree with 3.5× fewer cells reproduces it to four decimals, while a center-refined control (coarsest cells at the front) deviates by +2.35% — the ionization front is the one place needing resolution (Figure 6). With the thermal balance on, $T_\mathrm{e}(r)$ matches the 1D thermal reference to a median 0.14% and a mean bias +0.006% (Figure 7); without metals the metal-free nebula runs at 19–24 kK, the known overshoot that motivates the trace-metal cooling.

5.2. *The Lexington benchmarks*

The HII20 and HII40 cases of the Lexington/Meudon comparison (G. Ferland 1995; D. Péquignot et al. 2001) use 20 and 40 kK blackbody sources ($Q_\mathrm{H} = 1.0$ and $4.3 \times 10^{49}\,\mathrm{s}^{-1}$), $n_\mathrm{H} = 100$ in a hollow shell, and the standard Lexington abundances (He 0.1, C 2.2×10^{-4}, N 4×10^{-5}, O 3.3×10^{-4}, Ne 5×10^{-5}, S 9×10^{-6}). They are the natural reference for a new code because eight independent codes published results for the same two models, so the benchmark carries its own measure of how large a disagreement is significant. Tables 4 and 5 place MoCHII next to those eight, in the compilation of B. Ercolano et al. (2003), together with the Monte Carlo code of K. Wood et al. (2004) that was added to the same tables afterwards.

The MoCHII runs compared here are configured to match the published calculations wherever the setup is a choice rather than a physical difference. The ionizing luminosity is set from the benchmark blackbody luminosity so that Q_H reproduces the specified 4.265×10^{49} (HII40) and $1.004 \times 10^{49}\,\mathrm{s}^{-1}$ (HII20). Stellar photon energies are sampled continuously from the Planck spectrum with a Sobol sequence, and every photoionization cross section is evaluated at the sampled energy (Section 3.1); the 32-bin radiation array is only a diagnostic tally. The diffuse field is transported explicitly with case-A rates and the He I channels (Section 3.7), the treatment the published codes use, so that the comparison is not confounded by the transfer treatment. The recombination-line emissivities stay on the case-B tables, because the Lyman lines are optically thick in these models whatever the continuum treatment is; the two cases are separate switches (`par%case_ab` for the ionization, `par%line_case` for the line tables). The grid is a uniform level-6 octree (64^3 leaves) run with 2^{25} stellar

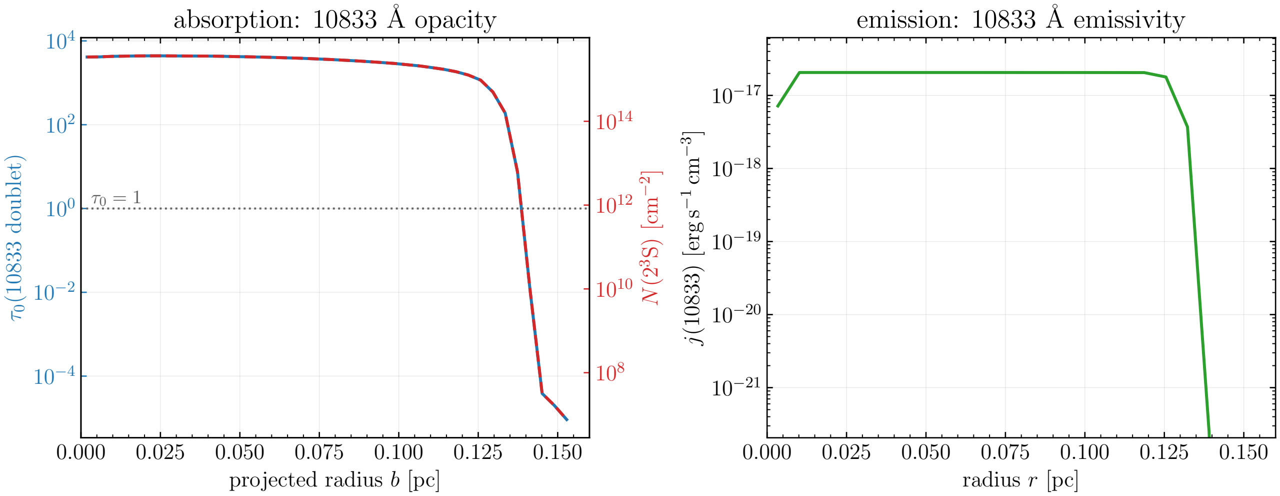

Figure 4. The He I 2^3S metastable diagnostic for a dense compact example ($n_{\mathrm{H}} = 10^4\ \mathrm{cm}^{-3}$, $R = 0.15$ pc, $T_{\mathrm{e}} = 10^4$ K), as radial profiles of the two sides of the same 2^3S–2^3P transition. Left, absorption: the 10833 Å line-center optical depth τ_0 of the blended $J' = 1, 2$ doublet against projected radius (left axis) with the 2^3S column density on the right axis. Both are line-of-sight integrals, and at the fixed T_{e} of this run the Doppler width is constant, so the two are strictly proportional and the right axis is the conversion of the left; the curves overlie. The region is optically thick inside $b = 0.139$ pc, where the optically-thin diagnostic breaks down. Right, emission: the 10833 Å emissivity against radius, from the R. L. Porter et al. (2012) He I block of the emissivity file.

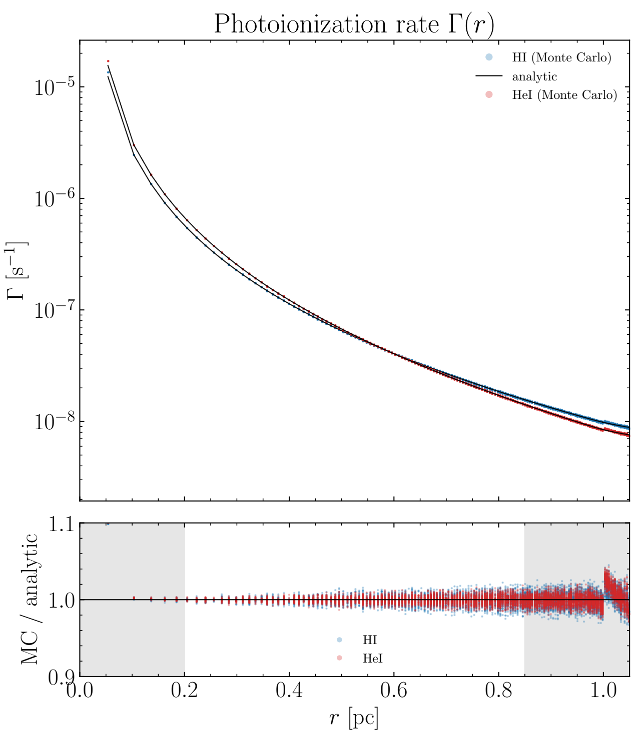

Figure 5. Rate-integral test: $\Gamma(r)$ from the Monte Carlo tally (points) against the analytic point-source attenuation (lines); the lower panel shows the ratio. Shaded regions are excluded (cell-size effects at the center, the voxelized sphere edge outside).

packets (Sobol seed 20260728) to the volume-integrated convergence criterion.

We also ran Cloudy c25.00 (C. M. Gunasekera et al. 2025) on both benchmarks and carry it as the C25 column of the same two tables, providing a uniform one-dimensional reference alongside the published multi-code spread. Its physics input is the unmodified regression deck that Cloudy distributes for each case, `tsuite/auto/hii_paris.in` for HII40 and `hii_coolstar.in` for HII20 — the same source, density, inner radius, and Lexington abundances as above — with only output commands appended; line luminosities are taken from a `save line list absolute` file and divided by the Cloudy Hβ luminosity, and the averaged temperature is Cloudy's own $n_{\mathrm{e}}n_{\mathrm{p}}$-weighted volume average, the same weighting used on both other sides. The one geometric difference is that each deck runs to its own stopping criterion rather than to a prescribed outer radius — an electron fraction of 10^{-2} at HII40, the 1000 K temperature floor at HII20 — so it carries a little more of the partly neutral edge than a model truncated at R_{out} would. The radii reached, 1.467×10^{19} and 8.97×10^{18} cm, sit at the top of the published R_{out} ranges (1.45–1.47×10^{19} and 8.7–9.01×10^{18} cm), so the two geometries are comparable, but the low-ionization rows fed by that edge — [O I], [S II], [N II] — are the ones where the difference could matter.

The temperature compared throughout this section is $\langle T[N_{\mathrm{p}}N_{\mathrm{e}}]\rangle$, the electron temperature averaged with the weight $n_{\mathrm{HII}}n_{\mathrm{e}}V$ — the same definition on both

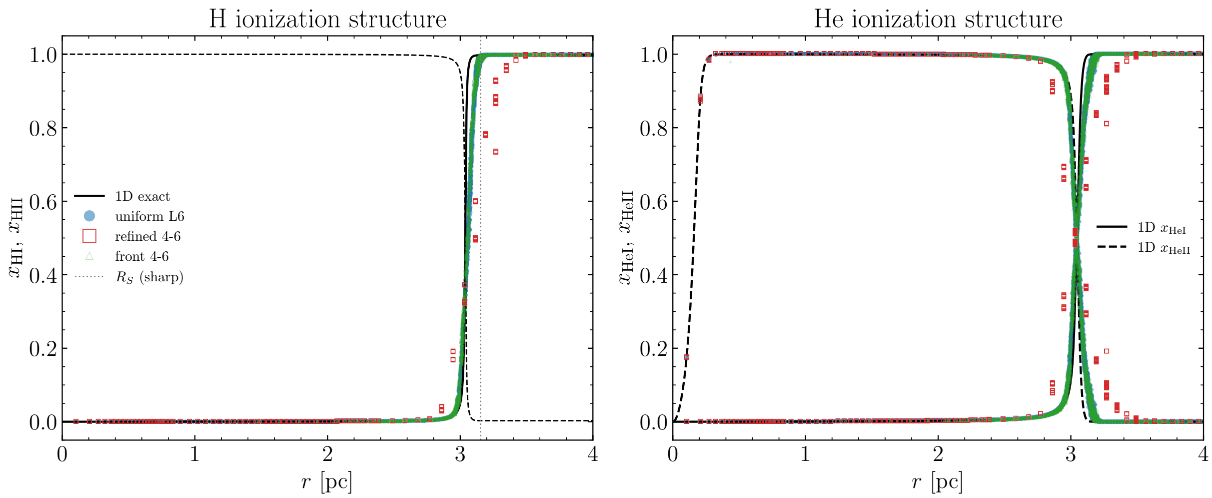

Figure 6. Strömgren test: H and He ionization structure against the 1D exact reference (black); uniform level-6 (blue), center-refined (red), and front-refined (green, overlying blue).

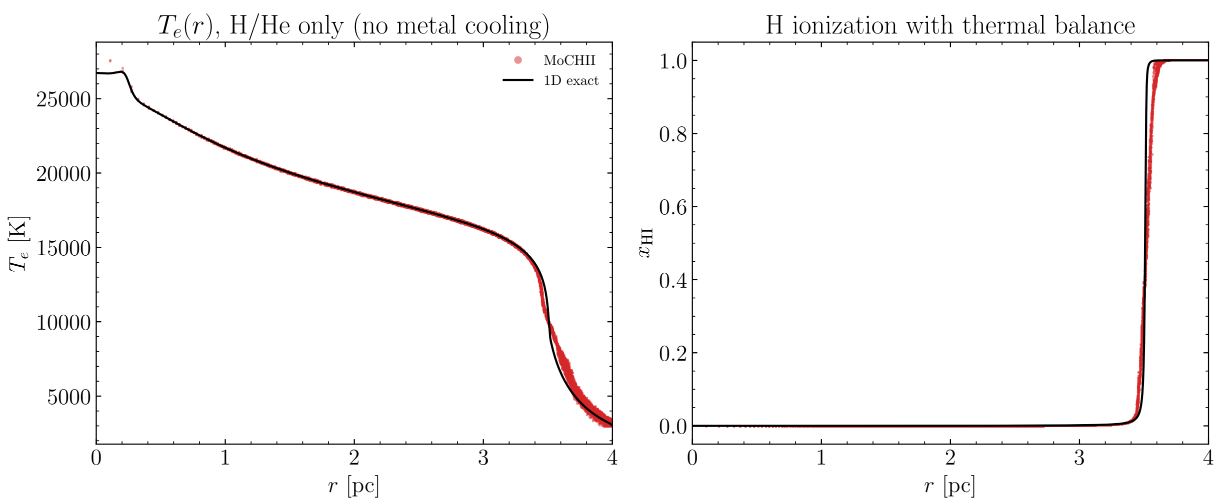

Figure 7. Thermal test: $T_{\rm e}(r)$ and $x_{\rm HI}(r)$ with the thermal balance, against the 1D exact reference.

sides, evaluated leaf by leaf in MoCHII and cell by cell in the published models. The inner-edge temperature $T_{\rm inner}$ that the published tables also list is quoted separately where it is used. Ionic fractions are $n_{\rm e}V$-weighted means, the convention of MOCASSIN's own output. As a further point of comparison we also ran MOCASSIN 2.02 itself, converged, on the HII40 model, and quote it below alongside the published columns.

Ionization structure. The helium and metal ionization is sensitive to the photon distribution around the He I edge. A grouped spectrum can assign all photons in a bin one representative energy and hence bias both the threshold budget and the opacity. The production calculations avoid that approximation: they sample the Planck probability density continuously and use the cross section at each sampled energy. The resulting $\langle{\rm He}^+\rangle/\langle{\rm H}^+\rangle$ is 0.703 at HII40, below the published 0.715–0.829 (median 0.767) and C25's 0.741, and 0.046 at HII20, near the published median of 0.048 inside a spread of 0.034–0.077.

Temperature. The weighted temperature is 8194 K at HII40 and 6916 K at HII20. Both match the Cloudy c25.00 solution of the same two models — 8210 K and 6910 K — to 0.2% and 0.1%, and both lie above the 2000-era published ranges (7720–8199 K, median 8026, and 6402–6749 K, median 6662). The inner-edge temperature is inside the published spread at both, 7538 K at HII40 against a published 7370–8318 K and 6938 K at HII20 against a published 6562–7224 K. The converged MOCASSIN 2.02 run of the HII40 model gives 8176 K, also inside the published spread. The two benchmark solutions therefore agree closely while remaining within the range set by the independent calculations.

The temperature is set by the cooling treatment, and the two limits of that treatment bracket the published set from opposite edges. The density-suppressed cooling used above — the level populations at the local n_e, Section 4.4 — gives the 8194 and 6916 K that track Cloudy; the zero-density limit of the same fits, retained as an option, overestimates the far-infrared line cooling at $n_e \sim 10^2\,\mathrm{cm}^{-3}$ and so places MoCHII instead at 7894 and 6376 K, inside and just below the published ranges but 4–8% below the two modern codes. The offset between the two limits agrees in sign and size with the far-infrared overestimate, which the next subsection measures directly.

5.3. *Cooling-budget verification at a fixed state*

An emission-line comparison cannot by itself separate an error in the atomic data from an error in the temperature, because each code's temperature is the root of its own energy balance: too little cooling in one channel raises T_e, which raises every other channel, and the converged line ratios absorb part of the discrepancy. The cooling data can be compared free of that feedback only at a *fixed* state. Cloudy c25.00 writes such a decomposition — its `save cooling each` output lists the cooling rate of every ion and every continuum channel zone by zone — so the test is to take its converged (T_e, n_e, ionic fractions) zone by zone, evaluate the MoCHII cooling terms on exactly that state, and integrate both over the model volume. What this measures is the absolute scale of two independent atomic-data compilations, channel by channel, with nothing else in either code entering. It is the comparison that isolates the cooling data from the temperature, which an equilibrium comparison cannot do, since the latter shows only the temperature at which each code's own heating and cooling cross.

On the HII40 model the two agree within the accuracy of the underlying collision data. With the n-level emissivities evaluated at the local n_e, the ratio MoCHII/Cloudy is 1.02 for [O III], 0.98 for [O II], 1.00 for [S III], 0.99 for [N II], 1.00 for [S II], 1.00 for [Ne II], 1.02 for [N III], 1.02 for [S IV], 0.99 for C III], 0.94 for [C II] and 0.99 for the H I collisional lines; summed over all line channels the two agree to 1.1%. Two channels fall outside ±10%, [Ne III] (1.30) and [O I] (0.84), and they carry 1.9% and 0.1% of the budget. The free-free continuum, an independent check of the Gaunt factor and of the net ion charges entering it, agrees to 0.2%. Recombination cooling differs by bookkeeping rather than by data: Cloudy reports the free-bound loss net of the escaping recombination continuum, $4.02 \times 10^{37}\,\mathrm{erg\,s^{-1}}$, which falls between the MoCHII case-A and case-B sums (6.61 and 3.62×10^{37}), the difference being the ground-state term that the explicit diffuse field re-emits instead of losing. The heating side of the same Cloudy model balances its cooling to 1.3×10^{-4} and deposits 4.57 eV for each ionizing photon, so the comparison state is itself an equilibrium one.

The same n-level emissivities at the local n_e are what the thermal loop uses by default (Section 4.4), so this fixed-state agreement is the cooling the converged benchmark carries. Replacing them by the zero-density limit of the same fits (Eq. 9, the `cooling_model = low_density` option) raises the summed line cooling to 1.116 times Cloudy's. The excess is not spread over the ions: it is 5.1 percentage points of the line budget in [O III] and 3.9 in [C II], and in both it sits entirely in the far-infrared fine-structure lines, whose critical densities are at or below the local $n_e = 110\,\mathrm{cm}^{-3}$. Since the line channels carry 79% of the cooling budget, that is a 9% excess in the total, and it accounts for the temperature difference quantitatively: scaling T_e on the fixed state until the MoCHII zero-density terms radiate the same total power — Cloudy balances its own heating at the unscaled temperature, and the photoheating of the two codes agrees — gives 7784 K against Cloudy's 8210 K, close to the 7894 K the converged zero-density run reaches. Removing the overestimate, the density-suppressed default reaches 8194 K, matching Cloudy to 0.2% (Section 5.2). The far-infrared overestimate, not the atomic data, is thus what separates the two limits, and the (T_e, n_e) table of Section 4.4 removes it.

Sensitivity to the other ingredients. The remaining modeling choices are smaller levers than the cooling function, and each is measured by an in-code experiment. The transport treatment is the largest of them: replacing the case-A explicit diffuse field by the case-B on-the-spot approximation, with every other setting fixed, gives 7898 K at HII40 and 6899 K at HII20, so the explicit field used here raises T_e by 4.0% and 1.8% and moves the two benchmarks the same way. Substituting MOCASSIN-style recombination (Péquignot 1991 radiative plus Nussbaumer–Storey 1983 dielectronic) for the Badnell 2023 data, whose $\alpha(\mathrm{X\,III} \to \mathrm{X\,II})$ are systematically lower (ratios 0.72–0.84), moves T_e by 0.15% and the carbon, nitrogen and oxygen ionization fractions by about 0.02, so the recombination vintage is a minor lever on both. The dominant [O III] collision strengths are stable across database generations: the ^{3}P–^{1}D effective collision strength is 2.283 in the CHIANTI v7 compilation behind the published MOCASSIN column and 2.262 in v11 at 10^4 K, a 0.9% difference worth less than 30 K. The photoionization heating is standard, with mean photoelectron energies $\langle h\nu - 13.6\rangle_{\rm H} = 3.9$ eV and $\langle h\nu - 24.6\rangle_{\rm He} = 4.1$ eV. Grid resolution is treated separately below.

Line ratios. The line intensities follow from that temperature and ionization structure rather than from the emission calculation, which is exact to the accuracy of its input data (Section 4.4: the n-level solver reproduces an independent reference solver to four digits, and the fitted collision data reproduce direct CHIANTI evaluations to 0.05–0.5%). At HII40 the infrared fine-structure lines, whose excitation energies are small enough that they are nearly temperature-independent, sit with the published set ([Ne III] 15.5 μm 0.283 in a published 0.263–0.429; [Ne II] 12.8 μm 0.206 in 0.177–0.217; [O III] 52+88 μm 2.43 just above 2.10–2.42), and so do the temperature-sensitive optical lines of the same ion: [O III] 5007+4959 is 2.36 against a published 1.64–3.27, and the auroral [O III] 4363, the most temperature-sensitive entry in the tables, is 0.0044 against a published 0.0022–0.0070. The lines that fall outside the published ranges are examined below.

Hβ and the photon budget. A recombination line is a photon counter, and $L(\mathrm{H}\beta)$ therefore tests whether the model is radiation-bounded: every ionizing photon absorbed by hydrogen returns as a recombination, and a fixed fraction of those recombinations produces Hβ. Both models here are bounded. At HII40 the ionization front is captured inside the prescribed sphere, at 1.45×10^{19} cm against an outer radius of 1.46×10^{19} cm, and the outermost shell is 57% neutral; at HII20 the outer shells are fully neutral. The budget closes accordingly: summing the case-B recombinations over the converged solutions gives $0.950\,Q_{\rm H}$ and $0.970\,Q_{\rm H}$. What hydrogen does not absorb is taken up by helium, which at HII40 absorbs $0.112\,Q_{\rm H}$ and returns most of it to hydrogen through the He I diffuse channels. The predicted $L(\mathrm{H}\beta) = 1.94 \times 10^{37}$ and 4.71×10^{36} erg s^{-1} are about 5% below the published medians (some 4% below the lowest published values), consistent with the 5.0% and 3.0% of $Q_{\rm H}$ that the recombination budget assigns to helium rather than to hydrogen, together with the weak temperature dependence of the effective Hβ emissivity at the higher benchmark temperature. The absolute Hβ luminosity is thus set by the recombination budget, and adds no discrepancy of its own.

The complete published line list. The full benchmark line lists, with all published codes, a CLOUDY c25.00 column, and the MOCHII column, are given in Appendix B (Tables 4 and 5); they repeat the comparison over the full benchmark line list in the form used by K. Wood et al. (2004), whose Tables 1 and 2 add their own Monte Carlo code (WME) to the eight other codes of the comparison; the MOCHII and C25 columns provide two additional reference solutions. The low-ionization edge line [O I] 6300+6363 is 0.0169 at HII40, above the published 0.0059–0.012 range; it is particularly sensitive to the thin partly neutral edge. The complete comparison is given directly in the tables.

The C25 column provides a uniform reference, and with the two temperatures matched the line ratios track C25 across the board. At HII40, where the weighted temperature equals C25's, both the temperature-insensitive far-infrared lines and the temperature-sensitive optical and auroral lines agree with C25 to a few percent: among the infrared lines [O III] 52+88 μm is 1.01× C25, [N III] 57.3 μm 1.01× and [S III] 18.7 μm 1.01×; among the optical ones [O III] 5007+4959 is 0.96×, its auroral 4363 0.96× and [O II] 3726+3729 1.00× — the same optical and auroral lines that the zero-density cooling, running 3.8% cooler, had suppressed to 0.72–0.84×. That the temperature-sensitive lines fall into line with C25 exactly when the temperature does is the line-level counterpart of the cooling match. At HII20, where MOCHII is 0.1% hotter than C25, [O II] 3726+3729 is 1.01× C25. The residual outliers at HII40 are not temperature: [Ne III] 15.5 μm (1.31×) and 3869+3968 (1.22×), [O I] 6300+6363 (1.51×, on a line the thin neutral edge and the grid resolution govern), and C III] 1907+1909 (0.90×), each an ionization or collision-strength difference in a single ion.

The C25 and published columns illustrate the benchmark's code-to-code spread. [S II] 6716+6731 at HII40 is the clearest case: the GF entry is 0.137, C25 gives 0.304 and MOCHII 0.322; the two latter values differ by 6%. Sulfur at HII20 is the same story in the other direction: [S III] 18.7 μm (0.215 against a published 0.34–0.46) and [S III] 9532+9069 (0.303 against 0.465–0.604) fall *below* every published code, and so does C25 (0.209 and 0.293), while [S II] 6716+6731 sits inside the published range for both (0.809 and 0.744). Both C25 and MOCHII have the same S II/S III ordering, which points at the sulfur data rather than at either code. A band experiment bounds one further difference: CLOUDY carries the full stellar spectrum, so photons below 13.6 eV ionize S^0 (10.36 eV) and C^0 (11.26 eV) in the partly neutral edge, while the benchmark band here starts at the hydrogen edge. Extending the MOCHII band down to 6 eV raises [S II] 6716+6731 by 6% at HII20 and by under 1% at HII40, with the other low-ionization lines moving by at most a few percent — a real ingredient of the low-ionization rows, but far too small to account for the [S III] difference.

A second, structural difference is worth naming here, because it affects sulfur specifically. The published columns are not all built on the same set of ionization stages: the code of K. Wood et al. (2004), and CMacIonize after it, omit neutral sulfur and start the sulfur chain at S^+, so that in gas where the ionizing field has been absorbed all sulfur is S^+ by construction; oxygen and neon in the same codes do carry their neutral stages. Mocassin instead solves the neutral stage of every element as an unknown, as MoCHII and Cloudy do. The convention matters only outside the ionization front, where it fixes the S^+ reservoir that feeds [S II] rather than letting it recombine to S^0; it is one of the ingredients of the scatter among the published sulfur entries, and it is worth keeping in view when a single [S II] ratio is used to judge a code.

Radial structure. Figures 14 and 15 give the radial ionization and temperature structure in the panel layout of K. Wood et al. (2004), their Figures 1 and 2, so that the two can be read panel by panel. The morphology is the published one throughout. At HII40 the hydrogen and helium neutral fractions rise monotonically from 4×10^{-5} and 2×10^{-4} at the inner edge, the singly and doubly ionized fractions of carbon, nitrogen and oxygen cross within 0.15 pc of one another near 4.0 pc, C^{+3} peaks at 0.05 and is confined to the inner region, and the temperature falls slightly out to about 2.2 pc before climbing steeply toward the front. At HII20 helium is essentially all neutral beyond 1.3 pc while hydrogen stays ionized out to 2.9 pc, so the doubly ionized metals occupy only a thin inner shell. Both models are radiation-bounded, so both outer radii are predictions. The ionized-volume-equivalent radius $(3V_{\rm ion}/4\pi)^{1/3}$ and the shell-averaged $x_{\rm HII} = 0.5$ crossing that the tables carry as $R_{\rm out}$ are 1.441 and 1.447×10^{19} cm at HII40 against a published 1.45–1.47×10^{19}, and 8.80 and 8.91×10^{18} cm at HII20 against a published 8.7–9.01×10^{18}: the two definitions of the outer boundary are not identical, but they locate the same front, and both fall inside the published range at HII20 and within 0.6% of its lower end at HII40, consistent with a recombination budget of 0.95–0.97 $Q_{\rm H}$. The temperature panels reproduce the published shape — a shallow inward decline followed by a steep rise at the front, where the surviving radiation field is hardest.

Grid resolution. The two three-dimensional codes of the comparison state what they used, and it is worth placing MoCHII beside them, because a Monte Carlo photoionization calculation on a Cartesian grid has a resolution as well as a photon budget. B. Ercolano et al. (2003) ran the published Mocassin benchmark columns on $13 \times 13 \times 13$ cells with the source in a *corner*, so that the grid covers one octant of the spherically symmetric model; in their words, "although the number of sampling points along each orthogonal axis is only 13, this is the equivalent of a one-dimensional code with 273 radial points." K. Wood et al. (2004) used 65^3 cells over the full box with the star at the center, which they describe as matching a 33^3 Mocassin octant, i.e. a cell width of "3% of the radius," and 10^8 photon packets. The MoCHII runs here use a uniform level-6 octree, $64^3 = 262\,144$ leaves over the full box with the source at the center, and 2^{25} packets. MoCHII and K. Wood et al. (2004) are therefore at essentially the same resolution — 4.6×10^{17} against 4.4×10^{17} cm on a side at HII40 — and both are about 2.5× finer per axis than the octant grid behind the published Mocassin column. We did not repeat the benchmark on a deliberately coarsened grid, because the quantity at issue is already resolution-converged in the direction that matters: the front re-refinement test of Section 6 shows $R_{\rm eff}$ changing by +0.008% between a native level-7 grid and a coarser re-refined equivalent, and B. Ercolano et al. (2003) themselves attribute to grid coarseness only their inner-edge temperature $T_{\rm inner}$ — which their coarse cells make too *low*, since a cell carries the temperature of its center rather than of the illuminated face — and not $\langle T[N_{\rm p}N_{\rm e}]\rangle$ or the line ratios compared here. Grid resolution is thus not a candidate for the differences discussed above, with the exception of the thin partly neutral edge that feeds [O I] 6300+6363, which one cell width does not resolve in any of the three-dimensional codes.

The radial ionization and temperature structure of both benchmarks is shown in Appendix B (Figures 14 and 15), in the panel layout of K. Wood et al. (2004).

Reference-code context. The C25 and GF columns demonstrate the intrinsic spread among implementations and atomic-data choices. For example, at HII40 [S II] 6716+6731 is 0.304 (C25) and 0.137 (GF), while [O III] 4363 is 0.0046 and 0.00235. At HII20 the two [S III] entries are 0.209 and 0.293 in C25, compared with 0.445 and 0.501 in GF. The published spread and the independently calculated C25 column therefore provide more informative reference ranges than any single column.

Where MoCHII *sits among the codes.* D. Péquignot et al. (2001) and B. Ercolano et al. (2003) note that the eight codes do not scatter randomly: GF, HN, and DP agree more tightly with one another — and, in the planetary-nebula cases, with Mocassin — than with TK, PH, and RS, a coherence they attribute to more recently updated atomic data and to a more careful treatment of the diffuse radiation field. A second grouping runs along the transfer treatment: the codes that solve the transfer exactly (RR, PH, and Mocassin) give systematically lower temperatures at the illuminated inner edge, because the radiation field there is softer than an approximate treatment makes it. MoCHII belongs with the first group on the data axis — exact hydrogenic and Verner cross sections, Badnell 2023 recombination, CHIANTI v11 collision strengths, the same vintage Cloudy uses — and with the second on the transfer axis, since the analytic direct-field estimator and the explicit diffuse field used for these benchmarks are an exact treatment in three dimensions; its inner-edge temperature is correspondingly in the lower half of the published $T_{\rm inner}$ range. Its ionization structure, its outer radius, its Hβ luminosity and most of its line ratios land inside the published spread, with helium at the low edge of it in the same direction as Cloudy c25.00, and its weighted temperature matches that solution to 0.2% at HII40 and 0.1% at HII20. The cooling treatment that achieves this, the density suppression of Section 4.4, is what separates it from the zero-density limit that leaves both benchmarks 4–8% cooler; the fixed-state comparison quantifies the offset between the two and shows it to be a property of the cooling function, not of the atomic data.

6. SOLUTION-DRIVEN FRONT REFINEMENT

A distinguishing capability of the octree grid is that the refinement can be driven by the ionization solution itself. At a chosen iteration the octree is rebuilt: a leaf is flagged as "front" when $\epsilon < x_{\rm HI} < 1-\epsilon$ or a face neighbor differs by more than a threshold; the new tree carries the maximum level wherever a cell overlaps a front-flagged leaf and a base level elsewhere. The gas state and opacity map over by position, and every dependent array is recreated. All ranks rebuild identically from the shared state, and the old grid's shared-memory windows are freed immediately after the position sampling, so repeated re-refinement neither leaks nor holds two grids in memory at once.

On the Strömgren configuration, starting at level 5 and rebuilding to level 7 at iteration 15 (32 768 $\to$ 464 311 leaves), the re-refined grid gives $R_{\rm eff} = 3.0432$ pc against a native uniform level-7 grid's 3.0430 pc (2 097 152 leaves) — agreement to +0.008% with 4.5$\times$ fewer cells, and both improve on level 6 (+0.56%), as resolution convergence expects (Figure 8). The front profiles overlie exactly.

As a demonstration on a realistic geometry, MoCHII was run on an Illustris-TNG 60-kpc cutout (density and neutral-fraction columns from the simulation), with a $Q_{\rm H} = 10^{53}\,{\rm s}^{-1}$, 40-kK source at the density peak and thermal balance, metals, and re-refinement enabled (Figure 9). Dense clumps cast sharp neutral shadows, the circumgalactic gas sits at $x_{\rm HI} \sim 10^{-4}$ with an $n_{\rm e}$-

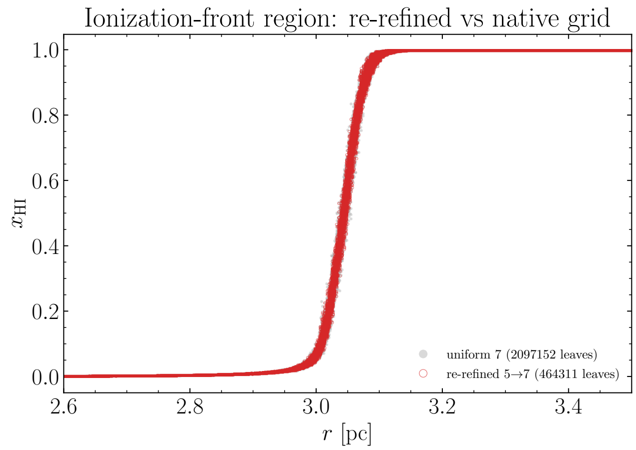

Figure 8. Re-refinement test: the front profiles of the re-refined 5 $\to$ 7 octree and the native uniform level-7 grid overlie exactly, at 4.5$\times$ fewer cells.

weighted mean $T_{\rm e} = 7502$ K, and the refinement (32 768 $\to$ 617 135 leaves) tracks the *distributed* ionization topology of a turbulent medium — clump skins and partially ionized gas rather than a single thin shell. This is the regime for which the volume-integrated convergence criterion (Section 3.6) was designed.

7. DUST AND PAH COUPLING

Because MoCHII is built on a dust radiative-transfer engine, the dust physics is a first-class part of the transport. The coupling has four parts.

Absorption competition. Each band bin adds a dust absorption opacity $\kappa_{\rm d} = \rho\kappa\, s_{\rm abs}(b)$ with $s_{\rm abs}(b) = (1 - a(E_b))\, C_{\rm ext}(E_b)/C_{\rm ext}(\lambda_{\rm ref})$ from a WD01/D03 $R_V = 3.1$ extinction table (J. C. Weingartner & B. T. Draine 2001; B. T. Draine 2003) covering the EUV, where $C_{\rm abs}/{\rm H} = 1.75 \times 10^{-21}$ at 13.6 eV falls to $3.96 \times 10^{-22}\,{\rm cm}^2$ at 100 eV. On the dusty Strömgren test, against a 1D reference carrying identical dust absorption, the full MW dust gives $R_{\rm eff} = 2.206$ pc versus the reference 2.198 pc (+0.34%), reproducing the classic dusty reduction $R/R_0 = 0.722$ (Figure 10); dust absorbs $\sim$60% of the band luminosity there.

Scattering. With scattering enabled the scattering opacity joins the extinction (in-band Henyey–Greenstein asymmetry $g = 0.67$–0.99, strongly forward), and interactions are sampled by a forced-first-flight scheme after the analytic direct walk, with an albedo weight at each interaction and Russian roulette. The full-scattering Strömgren radius, 2.197 pc, is within 0.4% of the absorption-only 2.206 pc: the in-band dust is strongly forward-throwing, so a scattered photon is barely deflected and scattering acts almost as pure transmission, exactly as expected.

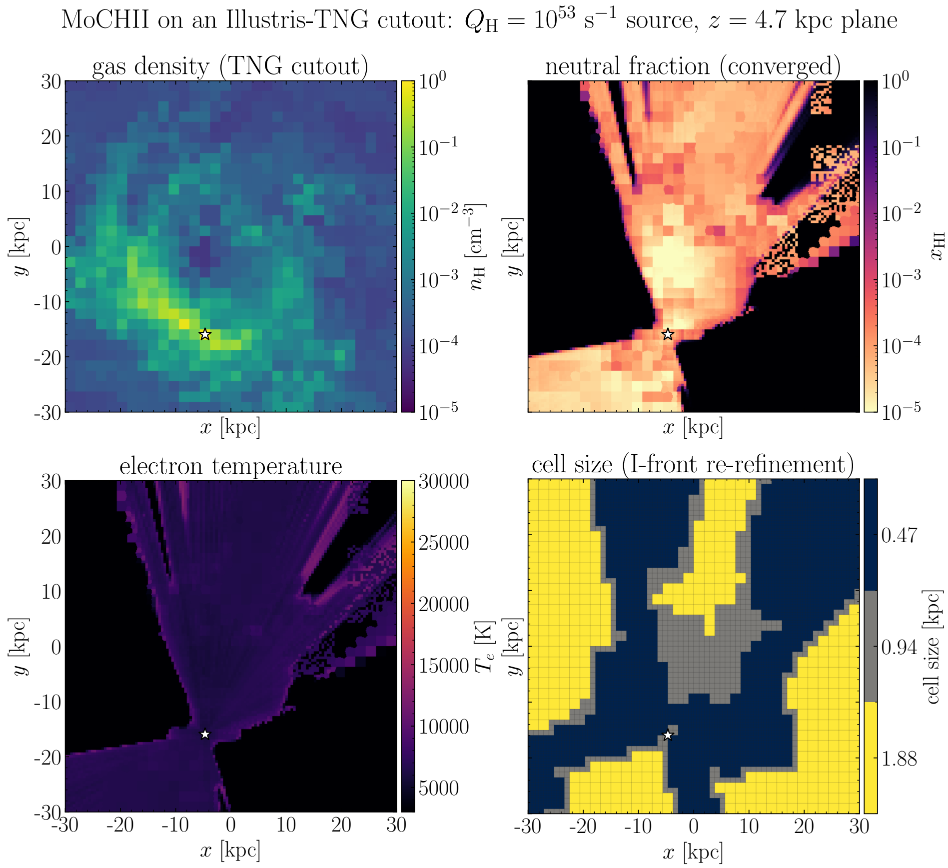

Figure 9. MoCHII on an Illustris-TNG cutout: gas density, converged neutral fraction (note the clump shadows), electron temperature, and the re-refined cell size, in the source plane.

Live ionization-dependent dust with a PAH split. Following P. Laursen et al. (2009), the dust content is traced by the metallicity-scaled hydrogen, with neutral gas carrying the full dust column and ionized gas only a survival fraction $f_{\rm ion}$ of it (grains are eroded where the hydrogen is ionized). MoCHII evaluates this on the *computed* neutral fraction and refreshes it every iteration, and it splits the grains into a large-grain share $1-f_{\rm PAH}$ and a PAH share $f_{\rm PAH}$ with independent ionized-gas survival, so the dust opacity of a leaf is

$$\kappa_{\rm d} \propto \frac{Z}{Z_{\rm ref}}\, n_{\rm H}\big[(1-f_{\rm PAH})(x_{\rm HI}+f_{\rm ion}\,x_{\rm HII}) + f_{\rm PAH}(x_{\rm HI}+f_{\rm ion,PAH}\,x_{\rm HII})\big], \quad (10)$$

with $x_{\rm HII}=1-x_{\rm HI}$ and $f_{\rm ion,PAH}=0$ by default (PAHs destroyed in ionized gas). Because this dust opacity is rebuilt from the ionization state each iteration while that state is in turn set by the dust-attenuated field, the dust–ionization feedback loop is solved self-consistently; it converges to a near dust-free interior ($R/R_0 = 0.996$).

Figure 10 contrasts the three treatments on the same Strömgren configuration. Static full MW dust competes with hydrogen for the ionizing photons throughout the volume, so the ionization front is pushed inward to $R_{\rm eff} = 2.21$ pc, the classic dust-limited size ($R/R_0 = 0.722$). Under `laursen09_live`, by contrast, the interior dust vanishes together with the neutral fraction it is tied to — once the gas is ionized it carries essentially no dust to absorb the ionizing field — so the front profile overlies the dust-free case almost exactly ($R_{\rm eff} = 3.04$ pc, $R/R_0 = 0.996$). The physical distinction is that only the neutral gas beyond the front and its thin skin retain dust, which is where the dust column that matters for the escaping radiation actually sits.

Grain heating, equilibrium temperature, and IR emission. The transported grain heating drives the equilibrium dust temperature by balancing the absorbed power against $4\pi\int s_{\rm abs}(\nu)B_\nu(T)\,d\nu$; with the band widened to 6 eV, $T_{\rm d}$ runs from 87 K at 0.3 pc to 28 K at the sphere edge (the canonical H II-region gradient), and the IR spectrum integral closes on the absorbed power to $L_{\rm IR}/L_{\rm abs} = 1.000$. A full stochastic dust emission solve (the MoCafe SEDust module; astrodust, DL07, and Zubko grain models with PAH bands, after B. T. Draine & A. Li 2007) is available at output time: it uses the size-resolved stochastic-heating spectrum as a *shape* nor-

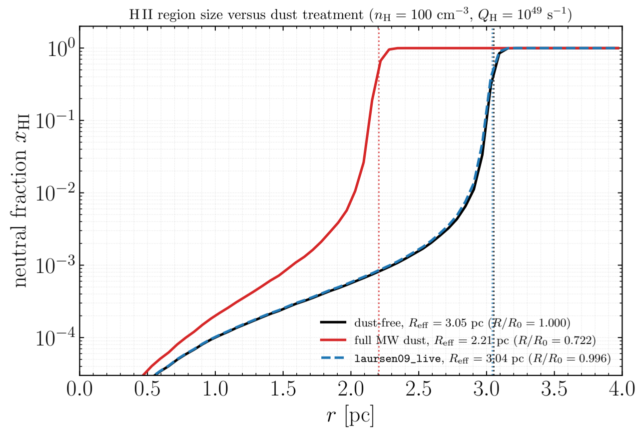

Figure 10. H II region size versus dust treatment: the radial neutral-fraction profile $x_{\rm HI}(r)$ for the dust-free, full MW dust, and `laursen09_live` runs of the same Strömgren configuration ($n_{\rm H} = 100\,{\rm cm}^{-3}$, $Q_{\rm H} = 10^{49}\,{\rm s}^{-1}$, $T_\star = 4 \times 10^4$ K, case B). Dotted vertical lines mark each effective ionized radius $R_{\rm eff}$. Full MW dust pushes the ionization front inward to $R/R_0 = 0.722$, while the live ionization-dependent dust vanishes in the ionized interior and recovers the dust-free size ($R/R_0 = 0.996$).

malized to the transported absorbed power, so the normalization is exact by construction. With the live PAH share (Figure 11), the PAH bands drop to 0.15–0.26 of the PAH-everywhere reference and the projected $8\,\mu$m map turns from centrally peaked into a *limb-brightened annulus* just inside the ionization front — the classic Spitzer/JWST bubble morphology.

8. THE PHOTODISSOCIATION-REGION SKIN

With the FUV band extension and three optional switches (all off by default), MoCHII models the atomic photodissociation-region (PDR) skin just outside the ionization front. The switches feed the metal-cascade electrons into the $n_{\rm e}$ closure, add metal photoheating to the thermal balance, and add grain photoelectric heating and grain recombination cooling with the E. L. O. Bakes & A. G. G. M. Tielens (1994) fits driven by the local FUV field G_0, together with the H-impact fine-structure cooling of [C II] $158\,\mu$m and [O I] $63\,\mu$m (Ğ. Barinovs et al. 2005). The last is essential: the standard cooling fits are electron-impact only, and with $n_{\rm HI}/n_{\rm e} \sim 10^3$–$10^4$ in the PDR the hydrogen collisions are the coolant. Two guards keep the photoelectric heating physical (evaluated at $\min(T, 10^4\,{\rm K})$ so the fits stay in range, and weighted by $x_{\rm HI}$ so ionized gas, whose grains are EUV-charged and whose energy is already routed to the IR, is not overheated). Both are necessary: without them the fits are evaluated outside their validity range and the zone runs away to a false 12–35 kK.

Figure 13 shows the converged radial structure of a PDR test (a $n_{\rm H} = 100$ octree sphere with a central source and all three switches on). The interior is fully ionized with a sharp front; beyond the front, the C II fraction is unity, the electron density falls to the carbon-dominated value $n_{\rm e} \simeq A_{\rm C} n_{\rm H}$, and the electron temperature settles to a genuine PDR equilibrium (a ∼340 K to ∼208 K gradient, not pinned at the temperature floor). This is the atomic PDR skin only: there is no H_2/CO chemistry or line transfer, so the PDR temperatures are first-order estimates, adequate for the ion structure and rough [C II] emissivities.

9. SUMMARY AND OUTLOOK

MoCHII is a three-dimensional Monte Carlo photoionization code for dusty nebulae on adaptive octree grids, built by adding a self-consistent gas-physics layer to a validated dust radiative-transfer engine. Its design choices — octree AMR with solution-driven front refinement, a zero-variance analytic direct-field estimator with a dedicated Cartesian DDA backend, an explicit optional Monte Carlo diffuse field, a data-driven trace-metal registry with modern atomic data, first-class dust absorption/scattering and equilibrium/stochastic emission with PAH physics, and peel-off synthetic imaging — distinguish it from the uniform-grid and 1D codes. It reproduces analytic attenuation to +0.004%, the Strömgren sphere to +0.56%, and a thermal H/He nebula to 0.14%. On the Lexington 2000 H II region benchmarks, with continuously sampled Planck photon energies and cross sections evaluated at each sampled energy, and with line cooling taken from the level populations at the local electron density, the $n_{\rm p} n_{\rm e}$-weighted temperature is 8194 K at HII40 and 6916 K at HII20, matching the CLOUDY c25.00 solution of the same two models (8210 and 6910 K) to 0.2% and 0.1%; both models are radiation-bounded, so both outer radii are predicted, and $L({\rm H}\beta)$ follows the recombination budget to within 8%. The line-cooling data are checked independently of the temperature by evaluating the MoCHII cooling terms on a fixed state taken from a CLOUDY c25.00 cooling decomposition: channel by channel the two compilations agree, and the summed line cooling agrees to 1%. The zero-density limit of the same cooling data, retained as an option, overestimates the far-infrared line cooling by 12% at $n_{\rm e} = 110\,{\rm cm}^{-3}$ and places both benchmarks 4–8% cooler (7894 and 6376 K, inside and just under the published ranges); the density suppression that removes it is the default. The CLOUDY c25.00 column provides a uniform reference for both tables and has the same S II/S III ordering as MoCHII. The solution-driven re-refinement reproduces a native high-resolution

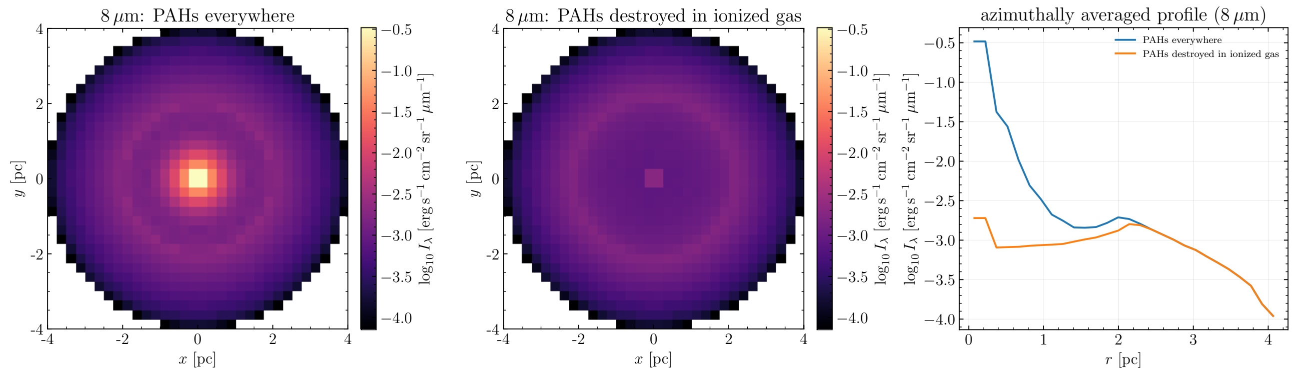

Figure 11. Stochastic dust emission at 8 μm with the ionization-dependent PAH share. The first two panels are two-dimensional projection maps sharing one brightness scale: the band emissivity, which MoCHII writes as the power radiated per unit volume and per unit wavelength into 4π sr, integrated along the line of sight and divided by 4π, so the panels show the specific intensity I_λ of an infrared-thin nebula. Left: PAHs surviving everywhere, giving a centrally peaked image. Middle: PAHs destroyed in the ionized interior ($x_{\rm HII}$-weighted survival). Right: the azimuthally averaged radial profiles of both maps. Destroying the PAHs removes the central peak and leaves a broad limb-brightened annulus just inside the ionization front at $r \simeq 2.1$ pc, visible in the profile as a local maximum a factor 1.9 above the depressed interior.

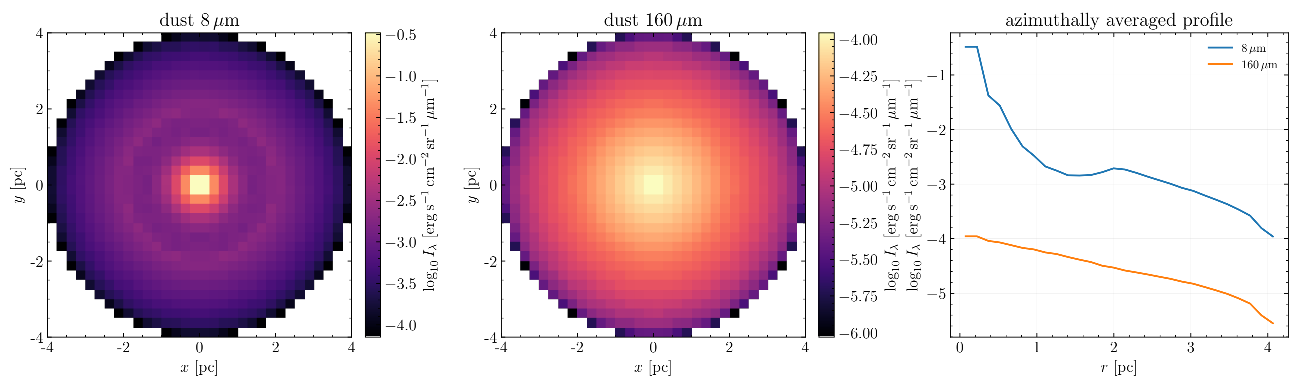

Figure 12. Dust emission at 8 and 160 μm from the stochastic SEDust solve, as two-dimensional projection maps of the specific intensity I_λ (left, middle) with their azimuthally averaged radial profiles (right). The band emissivity is written as the power radiated per unit volume and per unit wavelength into 4π sr, so I_λ is that emissivity integrated along the line of sight and divided by 4π. The short wavelength is centrally peaked (warm inner dust), falling by 3.5 dex across the sphere, while the far-infrared profile is far flatter (1.6 dex), the projected signature of the dust temperature gradient.

grid to +0.008% with 4.5× fewer cells, the dust coupling closes energy exactly and reproduces the limb-brightened Spitzer/JWST PAH morphology, and the atomic PDR skin recovers the carbon-dominated electron density and a physical PDR temperature.

Development is ongoing along several lines. A plane-parallel (xy-periodic slab) illumination mode with an emergent $I(\mu)$ boundary tally extends the code to slab and grazing-illumination geometries. A He I 2^3S metastable diagnostic supports synthetic 10833 Å absorption modeling. A hard-spectrum extension for planetary nebulae and active galactic nuclei — higher photon energies, extended ion stages, and inner-shell/Auger processes — is planned; the density-suppressed cooling introduced here already covers the high electron densities of planetary nebulae. The trace-metal registry makes further ions a data operation as the science requires them. Together these make MoCHII a tool for modern three-dimensional synthetic observations of H II regions, planetary nebulae, and the illuminated interstellar and circumgalactic medium.

ACKNOWLEDGMENTS

K.-I. Seon was supported by the Korea Astronomy and Space Science Institute under the R&D program (Project No. 2026-1-831-02) supervised by the Korea AeroSpace Administration.

Software: MoCHII (https://github.com/seoncafe/MoCHII), MoCafe (https://github.com/seoncafe/MoCafe), CHIANTI (K. P. Dere et al. 1997; R. P.

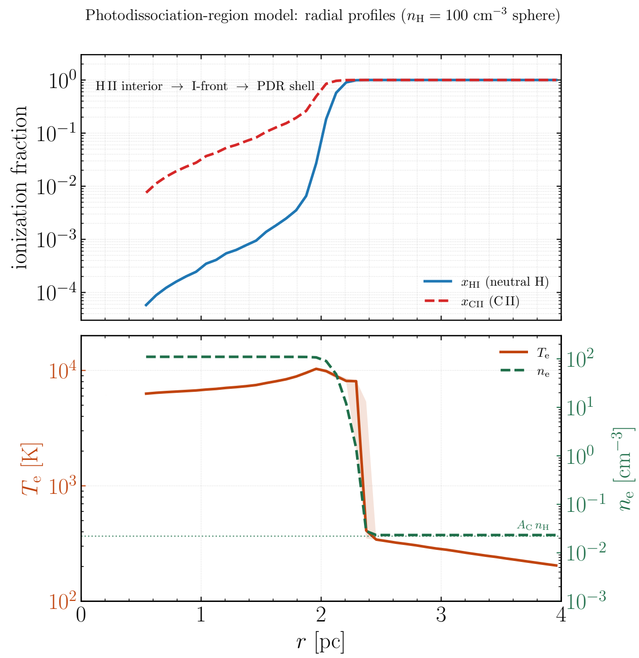

Figure 13. Radial structure of the PDR test: the neutral fraction, electron temperature, and electron density as a function of radius. Inside the front the gas is fully ionized; beyond it the electron density drops to the carbon-dominated value and the temperature settles to a PDR equilibrium above the floor.

Dufresne et al. 2024), MOCASSIN (B. Ercolano et al. 2003), Cloudy (M. Chatzikos et al. 2023), astropy, matplotlib, NumPy

APPENDIX

A. FITTED ATOMIC-DATA COEFFICIENTS

The atomic data are fitted offline and evaluated online (Section 4.4). Table 2 gives the quality of the line-cooling fits of every ion and Table 3 summarizes the n-level diagnostic models; it is generated mechanically from the data files by `paper/tables/make_atomic_tables.py`. The coefficients themselves — the amplitudes and excitation temperatures of the line-cooling fits, and the Burgess–Tully Chebyshev $\Upsilon(T)$ sets of the n-level models, several for each transition — are distributed with the code rather than reproduced here.[5]

[5] The fitted line-cooling coefficients of every ion are provided as text in `data/atomic/cooling_fit_parameters.txt` (and, one file for each ion, in `data/atomic/cooling_<ion>.txt`), with the fit form, units, fit range, and maximum fit error documented in `data/atomic/README.md`; the n-level models are in `data/atomic/nlevel_<ion>.txt`. Every file carries a provenance header.

Table 2. Quality of the collisional line-cooling fits: the number of exponential terms N, the maximum fit error of each ion over the full 10^3–10^5 K range, and the maximum error restricted to the temperatures that carry the cooling ($\Lambda > 10^{-3}\Lambda_{\rm max}$). The two agree for most ions; where they differ, the large full-range error sits in a temperature tail in which the ion radiates negligibly. The fitted coefficients themselves are distributed with the code (Appendix A).

Ion	N	max err	where it cools	Ion	N	max err	where it cools
H I	4	0.52%	0.52%	S IV	3	0.61%	0.61%
C I	6	0.27%	0.27%	Ar III	2	3.90%	3.90%
C II	5	0.35%	0.35%	Ar IV	4	0.27%	0.27%
C III	6	5.48%	0.50%	Mg II	4	0.22%	0.12%
N I	5	0.58%	0.34%	Fe II	5	0.56%	0.56%
N II	5	0.66%	0.66%	Fe III	5	0.48%	0.48%
N III	5	0.70%	0.70%	Si II	6	0.91%	0.91%
O I	6	0.20%	0.20%	Si III	6	27.70%	11.29%
O II	4	0.27%	0.15%	Si IV	2	0.90%	0.90%
O III	5	0.32%	0.32%	Cl II	3	8.05%	8.05%
Ne II	5	0.40%	0.40%	Cl III	6	9.15%	9.15%
Ne III	5	0.34%	0.34%	Cl IV	2	7.30%	7.30%
S II	3	2.23%	2.16%	Ca II	4	1.03%	0.42%
S III	5	0.23%	0.23%	Ca V	2	4.34%	4.34%

Table 3. n-level diagnostic models: number of levels, number of fitted effective-collision-strength transitions, and the worst transition fit error. The level energies and radiative A-values are from CHIANTI v11.0.2, and $\Upsilon(T)$ is a low-order Chebyshev fit in Burgess–Tully scaled space over 10^3–10^5 K; the full coefficient sets are in the provenance-headed files `data/atomic/nlevel_<ion>.txt`.

Ion	levels	transitions	worst err	Ion	levels	transitions	worst err
H I	25	45	0.20%	S IV	5	10	0.18%
C I	5	10	0.52%	Ar III	5	10	0.19%
C II	5	10	0.56%	Ar IV	5	10	0.20%
C III	5	10	0.35%	Mg II	3	3	0.15%
N I	5	10	0.78%	Fe II	13	78	0.20%
N II	5	10	0.17%	Fe III	14	91	0.20%
N III	5	10	0.45%	Si II	5	10	0.19%
O I	5	10	0.17%	Si III	5	10	0.65%
O II	5	10	0.27%	Si IV	3	3	0.54%
O III	6	15	1.68%	Cl II	5	10	0.17%
Ne II	2	1	0.16%	Cl III	5	10	0.19%
Ne III	5	10	0.19%	Cl IV	5	10	0.19%
S II	5	10	1.25%	Ca II	5	10	0.19%
S III	6	15	0.29%	Ca V	5	10	0.19%

Table 3 *continued*

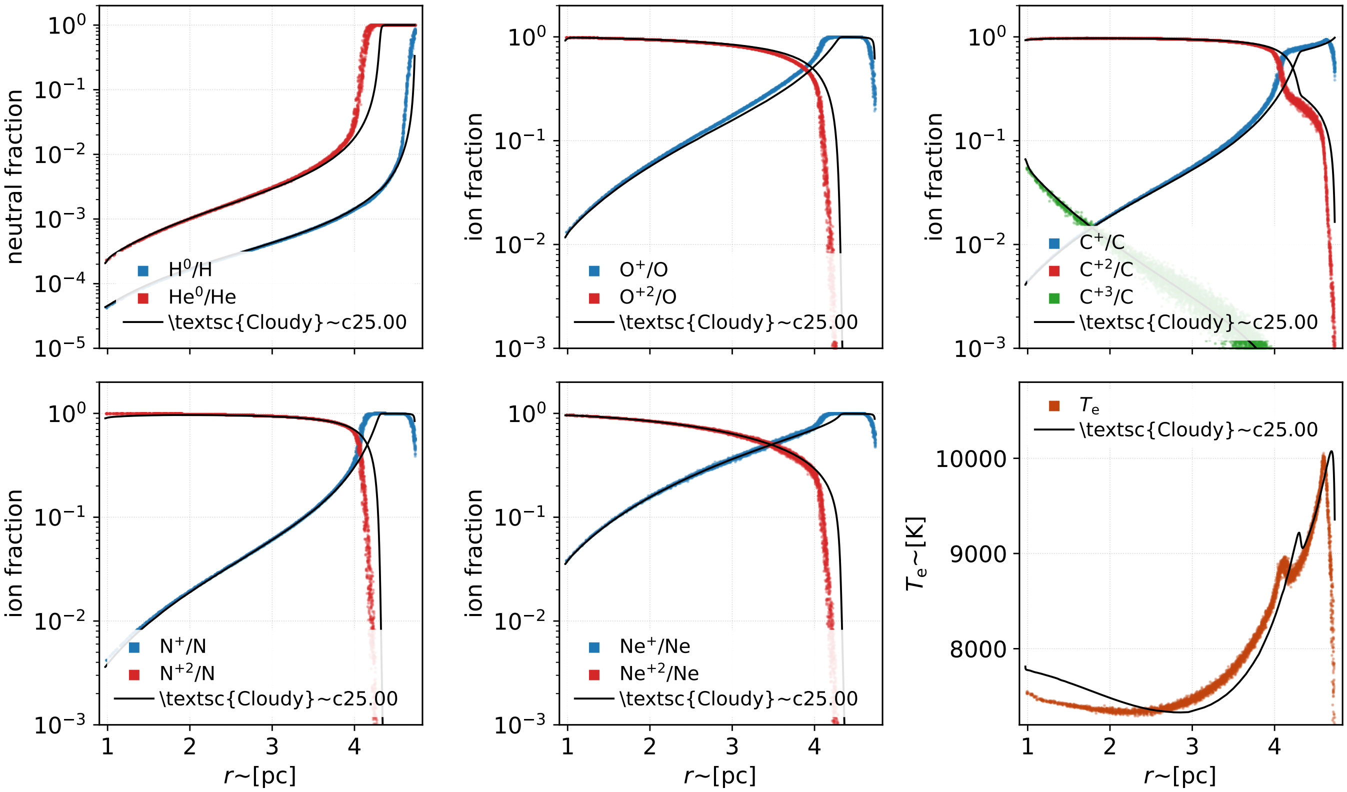

Figure 14. Lexington HII40 benchmark: radial ionization and temperature structure of the MoCHII solution, in the panel layout of K. Wood et al. (2004), their Figure 1. Each point is one octree leaf inside the prescribed shell, thinned at random for legibility; the temperature panel shows the leaves that are less than 95% neutral, the convention of K. Wood et al. (2004). Nitrogen stops at N^{+2} because the registry carries three nitrogen stages for this model. This is the same run as the MoCHII column of Table 4, with continuous Planck-sampled stellar photon energies and cross sections evaluated at the sampled energy; the 32-bin radiation array is diagnostic only. Black lines are CLOUDY c25.00 on the same model — the run behind the C25 column of Table 4, whose input is the unmodified c25.00 test-suite deck `tsuite/auto/hii_paris.in` — drawn as a line as K. Wood et al. (2004) draw CLOUDY in their own figures. The CLOUDY profile is cut at the outer radius of the MoCHII shell so that both codes cover one volume, and the 95% cut is applied to its zones as well. In the temperature panel the MoCHII points run below the CLOUDY curve through the interior and cross it at the front.

Table 3 *(continued)*

Ion	levels	transitions	worst err	Ion	levels	transitions	worst err

NOTE—Expanded 16/34-level Fe II/III models (`nlevel_fe_{2,3}_full.txt`) are available for iron-line studies. Ar II, Cl V, and Ca III/IV have no CHIANTI v11 level data and are tracked in the ionization balance without lines.

B. FULL LEXINGTON BENCHMARK LINE LISTS

Tables 4 and 5 give the complete Lexington HII40 and HII20 benchmark line lists discussed in Section 5.2, in the form used by K. Wood et al. (2004): the eight published codes of the B. Ercolano et al. (2003) compilation, the K. Wood et al. (2004) Monte Carlo code (WME), a CLOUDY c25.00 calculation (C25), and the MoCHII column. Line intensities are relative to Hβ.

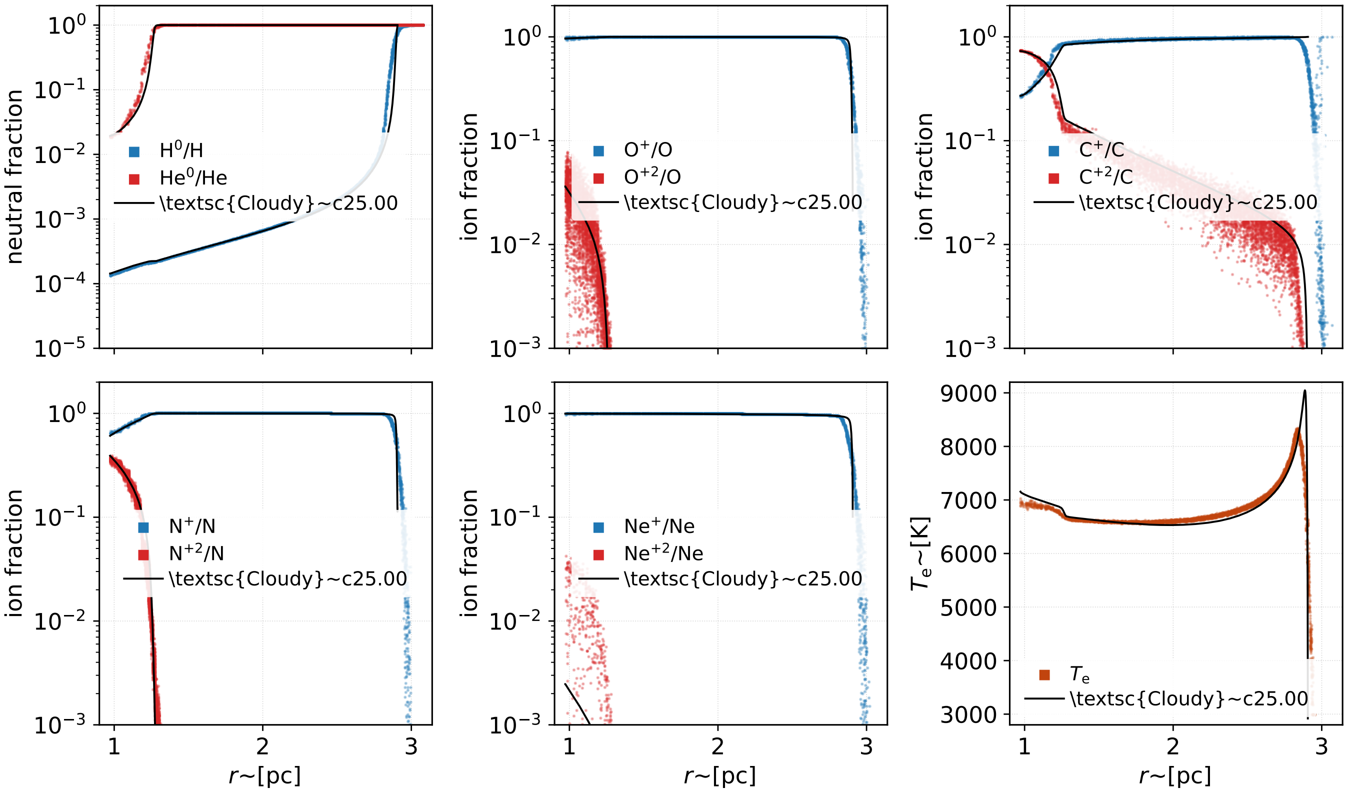

Figure 15. Lexington HII20 benchmark, in the panel layout of K. Wood et al. (2004), their Figure 2; conventions as in Figure 14. Ne^{+2} is confined within 1.4 pc, and C^{+3} stays below the plotted range and carries no series. This is the same run as the MoCHII column of Table 5. At 20 kK the helium-ionizing photons are so few that helium is neutral beyond 1.4 pc. The continuous-energy treatment nonetheless preserves the weak C^{+2} tail outside that helium transition. Black lines are Cloudy c25.00 on the same model, from the unmodified c25.00 deck `tsuite/auto/hii_coolstar.in` behind the C25 column of Table 5. Cloudy stops on its own criterion at 2.91 pc, short of the outer radius of the MoCHII shell, so its curves end there, and its final near-vertical fall is the recombination front itself, resolved in zones far finer than a MoCHII cell.

Table 4. Lexington H II region benchmark HII40 (40 kK blackbody): the published multi-code results of K. Wood et al. (2004, their Table 1), with a CLOUDY c25.00 calculation and the MoCHII result appended. Line intensities are relative to Hβ.

Quantity	GF	C25	HN	DP	TK	PH	RS	RR	BE	WME	MoCHII
$L(\mathrm{H}\beta)/10^{37}$ erg s^{-1}	2.06	2.06	2.02	2.02	2.10	2.05	2.07	2.05	2.02	2.01	1.94
He I 5876	0.119	0.100	0.112	0.113	0.116	0.118	0.116	···	0.114	0.114	0.105
C II] 2325+	0.157	0.189	0.141	0.139	0.110	0.166	0.096	0.178	0.148	0.172	0.173
C III] 1907+1909	0.071	0.062	0.076	0.069	0.091	0.060	0.066	0.074	0.041	0.078	0.056
[N II] 122 μm	0.027	0.030	0.037	0.034	···	0.032	0.035	0.030	0.036	0.031	0.032
[N II] 6584+6548	0.669	0.669	0.817	0.725	0.69	0.736	0.723	0.807	0.852	0.742	0.701
[N II] 5755	0.0050	0.0060	0.0054	0.0050	···	0.0064	0.0050	0.0068	0.0061	0.0057	0.0060
[N III] 57.3 μm	0.306	0.286	0.261	0.311	···	0.292	0.273	0.301	0.223	0.308	0.288
[O I] 6300+6363	0.0094	0.0112	0.0086	0.0088	0.012	0.0059	0.0070	···	0.0065	0.011	0.0169
[O II] 7320+7330	0.029	0.033	0.030	0.031	0.023	0.032	0.024	0.036	0.025	0.033	0.035
[O II] 3726+3729	1.94	2.32	2.17	2.12	1.6	2.19	1.88	2.26	1.92	2.23	2.31
[O III] 52+88 μm	2.35	2.40	2.10	2.26	2.17	2.34	2.29	2.34	2.28	2.42	2.43
[O III] 5007+4959	2.21	2.46	2.38	2.20	3.27	1.93	2.17	2.08	1.64	2.34	2.36
[O III] 4363	0.00235	0.0046	0.0046	0.0041	0.0070	0.0032	0.0040	0.0035	0.0022	0.0044	0.0044
[Ne II] 12.8 μm	0.177	0.200	0.195	0.192	···	0.181	0.217	0.196	0.212	0.192	0.206
[Ne III] 15.5 μm	0.294	0.216	0.264	0.270	0.35	0.429	0.350	0.417	0.267	0.263	0.283
[Ne III] 3869+3968	0.084	0.058	0.087	0.071	0.092	0.087	0.083	0.086	0.053	0.063	0.071
[S II] 6716+6731	0.137	0.304	0.166	0.153	0.315	0.155	0.133	0.130	0.141	0.18	0.322
[S II] 4068+4076	0.0093	0.0142	0.0090	0.0100	0.026	0.0070	0.005	0.0060	0.0060	0.0094	0.0139
[S III] 18.7 μm	0.627	0.504	0.750	0.726	0.535	0.556	0.567	0.580	0.574	0.623	0.509
[S III] 9532+9069	1.13	1.00	1.19	1.16	1.25	1.23	1.25	1.28	1.21	1.44	1.01
$10^3 \times \Delta(\mathrm{BC}\,3645)/\text{Å}$	4.88	···	···	4.95	···	5.00	5.70	···	5.47	4.96	···
T_{inner} / K	7719	7812	7668	7663	8318	7440	7644	7399	7370	7743	7538
$\langle T[N_{\mathrm{p}}N_{\mathrm{e}}]\rangle$ / K	7940	8210	7936	8082	8199	8030	8022	8060	7720	8183	8194
$R_{\mathrm{out}}/10^{19}$ cm	1.46	1.47	1.46	1.46	1.45	1.46	1.47	1.46	1.46	1.45	1.45
$\langle \mathrm{He}^+\rangle/\langle \mathrm{H}^+\rangle$	0.787	0.741	0.727	0.754	0.77	0.764	0.804	0.829	0.715	0.771	0.703

NOTE—Codes, in the notation of K. Wood et al. (2004): GF = Ferland's CLOUDY; HN = Netzer's ION; DP = Péquignot's NEBU; TK = Kallman's XSTAR; PH = Harrington's code; RS = Sutherland's MAPPINGS; RR = Rubin's NEBULA; BE = Ercolano's MOCASSIN; WME = the code of K. Wood et al. (2004) itself. Columns GF–BE agree entry by entry with Tables 4 and 5 of B. Ercolano et al. (2003), the compilation of the published columns. C25 is CLOUDY c25.00 (C. M. Gunasekera et al. 2025) run for this paper and provides a uniform one-dimensional reference alongside the published multi-code spread. Its physics input is the unmodified c25.00 test-suite file `tsuite/auto/hii_paris.in` (HII40) or `hii_coolstar.in` (HII20) — CLOUDY's own maintained version of these benchmarks — with only output commands appended; line luminosities come from a `save line list absolute` file and are divided by the CLOUDY H I 4861.32 Å luminosity, and the CLOUDY blends summed for each row are listed in `tables/make_wood_tables.py`. Unlike the published columns, C25 runs to the stopping criterion of its own deck rather than to a prescribed outer radius — an electron fraction of 10^{-2} at HII40, the 1000 K temperature floor at HII20 — which mainly affects the low-ionization rows fed by the partly neutral edge. The MoCHII column uses continuous stellar photon energies sampled from the Planck spectrum with a Sobol sequence (seed 20260728), and evaluates photoionization cross sections at each sampled energy; the 32-bin ionizing array is a diagnostic tally only. It uses the explicit Monte Carlo diffuse field including the He I channels (case A, matching the transfer treatment of the published codes), and Badnell 2023 recombination with the Mao & Kaastra 2016 ground-level split, at 2^{25} stellar packets with the volume-integrated convergence criterion; the ionizing luminosity is set from the benchmark $L(\mathrm{BB})$ so that $Q(\mathrm{H})$ matches the specification; blends are the sum of the MoCHII components listed in `tables/make_wood_tables.py`, and MoCHII wavelengths are vacuum values. The MoCHII structural entries are computed from the same run: T_{inner} is the mean T_{e} of the cells at the illuminated inner edge, R_{out} the radius at which the shell-averaged x_{HII} falls through 0.5, and the helium ratio follows Eq. (17) of B. Ercolano et al. (2003), $\langle \mathrm{He}^+\rangle/\langle \mathrm{H}^+\rangle = \int n_{\mathrm{e}} n(\mathrm{He}^+)\,\mathrm{d}V / \int n_{\mathrm{e}} n_{\mathrm{p}}\,\mathrm{d}V \times n_{\mathrm{H}}/n_{\mathrm{He}}$. Where the ionized gas still fills the outermost shell of the prescribed sphere, R_{out} would be set by the grid rather than predicted, and the entry is left empty. The Balmer jump is not computed. The Hβ row is headed 10^{36} erg s^{-1} in the published table; the values listed there are the 10^{37} erg s^{-1} values of B. Ercolano et al. (2003), and the heading is corrected accordingly here.

Table 5. Lexington H II region benchmark HII20 (20 kK blackbody): the published multi-code results of K. Wood et al. (2004, their Table 2), with a Cloudy c25.00 calculation and the MoCHII result appended. Line intensities are relative to Hβ.

Quantity	GF	C25	HN	DP	TK	PH	RS	RR	BE	WME	MoCHII
$L(\mathrm{H}\beta)/10^{36}\,\mathrm{erg\,s^{-1}}$	4.85	4.99	4.85	4.83	4.90	4.93	5.04	4.89	4.97	4.87	4.71
He I 5876	0.0072	0.0061	0.008	0.0073	0.008	0.0074	0.0110	···	0.0069	0.00684	0.0068
C II] 2325+	0.054	0.076	0.047	0.046	0.040	0.060	0.038	0.063	0.042	0.053	0.068
[N II] 122 μm	0.068	0.070	···	0.072	0.007	0.072	0.071	0.071	0.071	0.067	0.073
[N II] 6584+6548	0.745	0.809	0.786	0.785	0.925	0.843	0.803	0.915	0.846	0.778	0.840
[N II] 5755	0.0028	0.0032	0.0024	0.0023	0.0029	0.0033	0.0030	0.0033	0.0025	0.0025	0.0032
[N III] 57.3 μm	0.0040	0.0038	0.0030	0.0032	0.0047	0.0031	0.0020	0.0022	0.0019	0.0049	0.0039
[O I] 6300+6363	0.0080	0.0084	0.0060	0.0063	0.0059	0.0047	0.0050	···	0.0088	0.055	0.0112
[O II] 7320+7330	0.0087	0.0113	0.0085	0.0089	0.0037	0.0103	0.0080	0.0100	0.0064	0.0089	0.0122
[O II] 3726+3729	1.01	1.40	1.13	1.10	1.04	1.22	1.08	1.17	0.909	1.12	1.41
[O III] 52+88 μm	0.0030	0.0033	0.0026	0.0026	0.0040	0.0037	0.0020	0.0017	0.0022	0.0044	0.0033
[O III] 5007+4959	0.0021	0.0023	0.0016	0.0015	0.0024	0.0014	0.0010	0.0010	0.0011	0.0026	0.0021
[Ne II] 12.8 μm	0.264	0.296	0.267	0.276	0.27	0.271	0.286	0.290	0.295	0.289	0.303
[S II] 6716+6731	0.499	0.744	0.473	0.459	1.02	0.555	0.435	0.492	0.486	0.521	0.778
[S II] 4068+4076	0.022	0.023	0.017	0.020	0.052	0.017	0.012	0.015	0.013	0.017	0.023
[S III] 18.7 μm	0.445	0.209	0.460	0.441	0.34	0.365	0.398	0.374	0.371	0.381	0.215
[S III] 9532+9069	0.501	0.293	0.480	0.465	0.56	0.549	0.604	0.551	0.526	0.602	0.303
$10^3 \times \Delta(\mathrm{BC}\,3645)/\text{Å}$	5.54	···	···	5.62	···	5.57	5.50	···	6.18	5.52	···
T_{inner} / K	7224	7155	6815	6789	6607	6742	6900	6708	6562	6852	6938
$\langle T[N_{\mathrm{p}}N_{\mathrm{e}}]\rangle$ / K	6680	6910	6650	6626	6662	6749	6663	6679	6402	6692	6916
$R_{\mathrm{out}}/10^{18}$ cm	8.89	8.97	8.88	8.88	8.7	8.95	9.01	8.92	8.89	8.73	8.89
$\langle \mathrm{He}^+\rangle/\langle \mathrm{H}^+\rangle$	0.048	0.043	0.051	0.049	0.048	0.044	0.077	0.034	0.041	0.047	0.046

Note—Codes, in the notation of K. Wood et al. (2004): GF = Ferland's Cloudy; HN = Netzer's Ion; DP = Péquignot's Nebu; TK = Kallman's XStar; PH = Harrington's code; RS = Sutherland's Mappings; RR = Rubin's Nebula; BE = Ercolano's Mocassin; WME = the code of K. Wood et al. (2004) itself. Columns GF–BE agree entry by entry with Tables 4 and 5 of B. Ercolano et al. (2003), the compilation of the published columns. C25 is Cloudy c25.00 (C. M. Gunasekera et al. 2025) run for this paper and provides a uniform one-dimensional reference alongside the published multi-code spread. Its physics input is the unmodified c25.00 test-suite file `tsuite/auto/hii_paris.in` (HII40) or `hii_coolstar.in` (HII20) — Cloudy's own maintained version of these benchmarks — with only output commands appended; line luminosities come from a `save line list absolute` file and are divided by the Cloudy H I 4861.32 Å luminosity, and the Cloudy blends summed for each row are listed in `tables/make_wood_tables.py`. Unlike the published columns, C25 runs to the stopping criterion of its own deck rather than to a prescribed outer radius — an electron fraction of 10^{-2} at HII40, the 1000 K temperature floor at HII20 — which mainly affects the low-ionization rows fed by the partly neutral edge. The MoCHII column uses continuous stellar photon energies sampled from the Planck spectrum with a Sobol sequence (seed 20260728), and evaluates photoionization cross sections at each sampled energy; the 32-bin ionizing array is a diagnostic tally only. It uses the explicit Monte Carlo diffuse field including the He I channels (case A, matching the transfer treatment of the published codes), and Badnell 2023 recombination with the Mao & Kaastra 2016 ground-level split, at 2^{25} stellar packets with the volume-integrated convergence criterion; the ionizing luminosity is set from the benchmark $L(\mathrm{BB})$ so that $Q(\mathrm{H})$ matches the specification; blends are the sum of the MoCHII components listed in `tables/make_wood_tables.py`, and MoCHII wavelengths are vacuum values. The MoCHII structural entries are computed from the same run: T_{inner} is the mean T_{e} of the cells at the illuminated inner edge, R_{out} the radius at which the shell-averaged x_{HII} falls through 0.5, and the helium ratio follows Eq. (17) of B. Ercolano et al. (2003), $\langle \mathrm{He}^+\rangle/\langle \mathrm{H}^+\rangle = \int n_{\mathrm{e}} n(\mathrm{He}^+)\,dV / \int n_{\mathrm{e}} n_{\mathrm{p}}\,dV \times n_{\mathrm{H}}/n_{\mathrm{He}}$. Where the ionized gas still fills the outermost shell of the prescribed sphere, R_{out} would be set by the grid rather than predicted, and the entry is left empty. The Balmer jump is not computed.

REFERENCES

Badnell, N. R. 2006, ApJS, 167, 334, doi: 10.1086/508465

Badnell, N. R., O'Mullane, M. G., Summers, H. P., et al. 2003, A&A, 406, 1151, doi: 10.1051/0004-6361:20030816

Bakes, E. L. O., & Tielens, A. G. G. M. 1994, ApJ, 427, 822, doi: 10.1086/174188

Barinovs, Ğ., van Hemert, M. C., Krems, R., & Dalgarno, A. 2005, ApJ, 620, 537, doi: 10.1086/426860

Burgess, A., & Tully, J. A. 1992, A&A, 254, 436

Burley, B. 2020, Journal of Computer Graphics Techniques, 9, 1

Chatzikos, M., Bianchi, S., Camilloni, F., et al. 2023, RMxAA, 59, 327, doi: 10.22201/ia.01851101p.2023.59.02.12

Dere, K. P. 2007, A&A, 466, 771, doi: 10.1051/0004-6361:20066728

Dere, K. P., Landi, E., Mason, H. E., Monsignori Fossi, B. C., & Young, P. R. 1997, A&AS, 125, 149, doi: 10.1051/aas:1997368

Draine, B. T. 2003, ARA&A, 41, 241, doi: 10.1146/annurev.astro.41.011802.094840

Draine, B. T. 2026, ApJ, 999, 3, doi: 10.3847/1538-4357/ae3171

Draine, B. T., & Li, A. 2007, ApJ, 657, 810, doi: 10.1086/511055

Dufresne, R. P., Del Zanna, G., Young, P. R., et al. 2024, ApJ, 974, 71, doi: 10.3847/1538-4357/ad6765

Ercolano, B., Barlow, M. J., & Storey, P. J. 2005, MNRAS, 362, 1038, doi: 10.1111/j.1365-2966.2005.09381.x

Ercolano, B., Barlow, M. J., Storey, P. J., & Liu, X.-W. 2003, MNRAS, 340, 1136, doi: 10.1046/j.1365-8711.2003.06371.x

Ercolano, B., & Storey, P. J. 2006, MNRAS, 372, 1875, doi: 10.1111/j.1365-2966.2006.10988.x

Ercolano, B., Young, P. R., Drake, J. J., & Raymond, J. C. 2008, ApJS, 175, 534, doi: 10.1086/524378

Ferland, G. 1995, in The Analysis of Emission Lines: A Meeting in Honor of the 70th Birthdays of D. E. Osterbrock & M. J. Seaton, ed. R. Williams & M. Livio, 83

Ferland, G. J., Chatzikos, M., Guzmán, F., et al. 2017, RMxAA, 53, 385, doi: 10.48550/arXiv.1705.10877

Gunasekera, C. M., van Hoof, P. A. M., Dehghanian, M., et al. 2025, RMxAA, 61, 120, doi: 10.22201/ia.01851101p.2025.61.03.01

Huang, C., Koskinen, T., Lavvas, P., & Fossati, L. 2023, ApJ, 951, 123, doi: 10.3847/1538-4357/accd5e

Hui, L., & Gnedin, N. Y. 1997, MNRAS, 292, 27, doi: 10.1093/mnras/292.1.27

Hummer, D. G. 1988, ApJ, 327, 477, doi: 10.1086/166210

Jin, Y., Kewley, L. J., & Sutherland, R. 2022a, ApJ, 927, 37, doi: 10.3847/1538-4357/ac48f3

Jin, Y., Kewley, L. J., & Sutherland, R. S. 2022b, ApJL, 934, L8, doi: 10.3847/2041-8213/ac80f3

Joe, S., & Kuo, F. Y. 2008, SIAM Journal on Scientific Computing, 30, 2635, doi: 10.1137/070709359

Kingdon, J. B., & Ferland, G. J. 1996, ApJS, 106, 205, doi: 10.1086/192335

Laursen, P., Sommer-Larsen, J., & Andersen, A. C. 2009, ApJ, 704, 1640, doi: 10.1088/0004-637X/704/2/1640

Lucy, L. B. 1999, A&A, 344, 282

Mao, J., & Kaastra, J. 2016, A&A, 587, A84, doi: 10.1051/0004-6361/201527568

Oklopčić, A., & Hirata, C. M. 2018, ApJL, 855, L11, doi: 10.3847/2041-8213/aaada9

Osterbrock, D. E., & Ferland, G. J. 2006, Astrophysics of gaseous nebulae and active galactic nuclei

Péquignot, D., Ferland, G., Netzer, H., et al. 2001, in Astronomical Society of the Pacific Conference Series, Vol. 247, Spectroscopic Challenges of Photoionized Plasmas, ed. G. Ferland & D. W. Savin, 533

Porter, R. L., Ferland, G. J., Storey, P. J., & Detisch, M. J. 2012, MNRAS, 425, L28, doi: 10.1111/j.1745-3933.2012.01300.x

Porter, R. L., Ferland, G. J., Storey, P. J., & Detisch, M. J. 2013, MNRAS, 433, L89, doi: 10.1093/mnrasl/slt049

Seon, K.-i. 2024, ApJ, 971, 184, doi: 10.3847/1538-4357/ad58bd

Seon, K.-i., & Kim, C.-G. 2020, ApJS, 250, 9, doi: 10.3847/1538-4365/aba2d6

Seon, K.-i., Song, H., & Chang, S.-J. 2022, ApJS, 259, 3, doi: 10.3847/1538-4365/ac3af1

Shull, J. M., & van Steenberg, M. E. 1985, ApJ, 298, 268, doi: 10.1086/163605

Storey, P. J., & Hummer, D. G. 1995, MNRAS, 272, 41, doi: 10.1093/mnras/272.1.41

Sutherland, R. S., & Dopita, M. A. 1993, ApJS, 88, 253, doi: 10.1086/191823

Sutherland, R. S., & Dopita, M. A. 2017, ApJS, 229, 34, doi: 10.3847/1538-4365/aa6541

van Hoof, P. A. M., Williams, R. J. R., Volk, K., et al. 2014, MNRAS, 444, 420, doi: 10.1093/mnras/stu1438

Vandenbroucke, B., & Wood, K. 2018, Astronomy and Computing, 23, 40, doi: 10.1016/j.ascom.2018.02.005

Verner, D. A., Ferland, G. J., Korista, K. T., & Yakovlev, D. G. 1996, ApJ, 465, 487, doi: 10.1086/177435

Verner, D. A., & Yakovlev, D. G. 1995, A&AS, 109, 125

Voronov, G. S. 1997, Atomic Data and Nuclear Data Tables, 65, 1, doi: 10.1006/adnd.1997.0732

Weingartner, J. C., & Draine, B. T. 2001, ApJ, 548, 296, doi: 10.1086/318651

Wood, K., Mathis, J. S., & Ercolano, B. 2004, MNRAS, 348, 1337, doi: 10.1111/j.1365-2966.2004.07458.x

Yan, D., Seon, K.-i., Guo, J., Chen, G., & Li, L. 2022, ApJ, 936, 177, doi: 10.3847/1538-4357/ac8793